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Contacting technology

Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

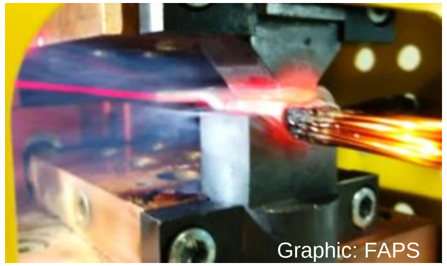
This lecture provides an overview of the physical principles of electrical contact and the most common contact technologies used in electromechanical engineering.

Contacting is one of the most complex and at the same time most critical process steps in the production of electric drives

After attending the lecture unit "Contacting technology" you should

- **understand the basic principles of electrical contacts,**
- **be able to name the different processes for removing electrical insulation,**
- **be able to differentiate the different methods of contacting as well as their advantages and disadvantages,**
- **be familiar with the methods and processes for testing electric contacts and**
- **be able to describe future challenges for contacting technology.**

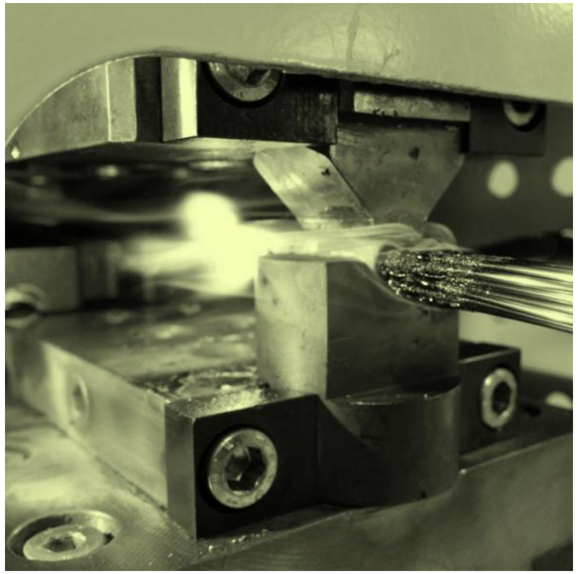

Graphic: FAPS


Graphic: FAPS


Graphic: Klaus Faber AG

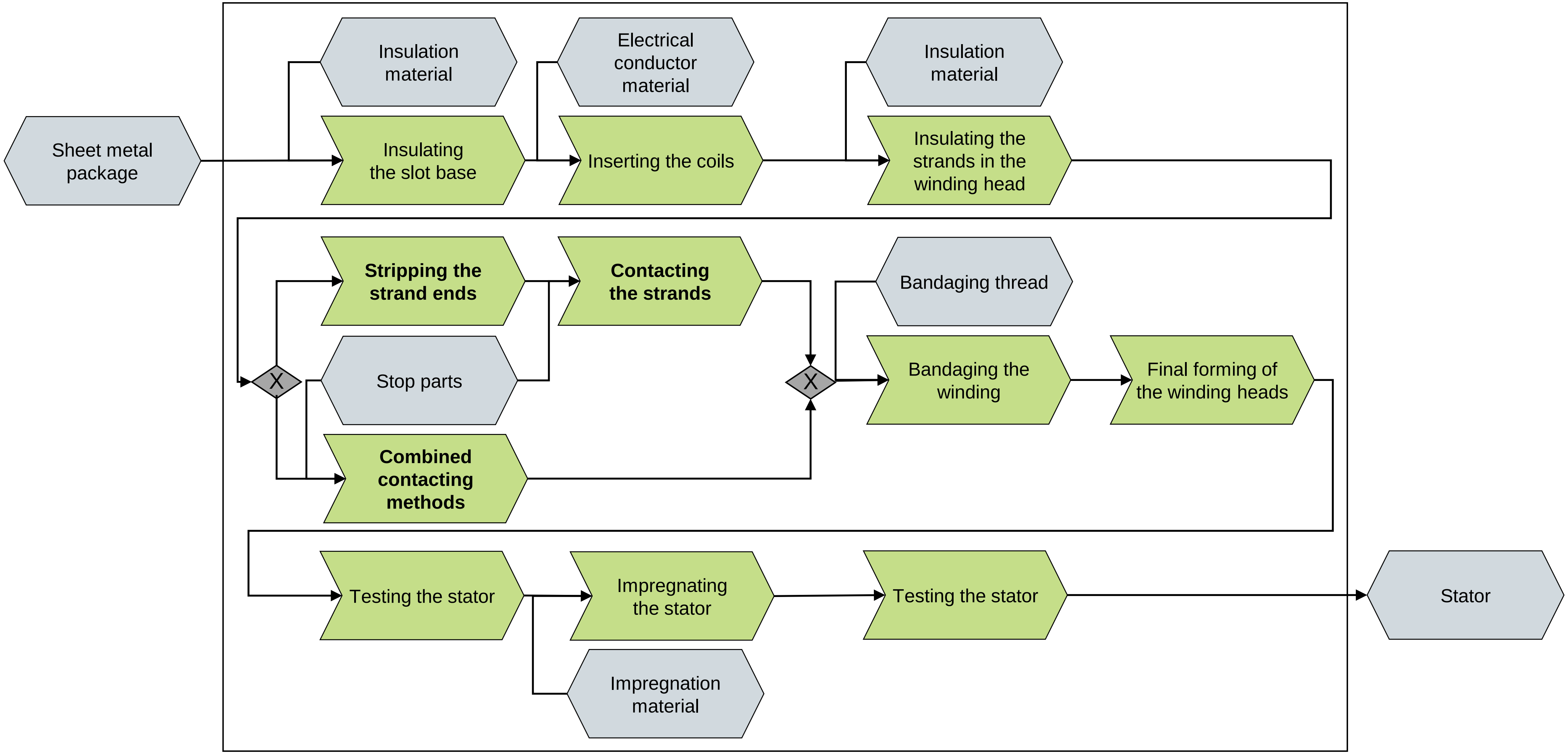
This lecture provides an overview of the physical principles of electrical contact and the most common contact technologies used in electromechanical engineering.

- **Contacting in the production of electric motors and machines**
- Basics of electrical contacts
- Insulation stripping process technologies
- Process technologies for the production of electrical contacts
- Testing of electrical contacts in electromechanical engineering
- Research activities in the field of contacting



Contacting is an essential process step in the production of electrical machines.

Slide from Lec. 6

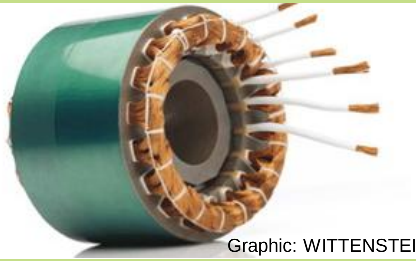


The contacting process has a significant influence on the lifetime of the electrical machine.

Contacting process:

- Winding ends of the stator are mechanically and electrically conductively connected to a contacting element
- It can be divided in the two essential process steps of stripping and joining

Winding ends of the stator



Graphic: WITTENSTEIN cyber motor GmbH

Terminal parts

Graphic: FAPS

Completed connection

Graphic: FAPS

Perspectives of contacting

Quality-related perspective

- Quality of the connection technology is decisive for the lifetime, especially in mobile applications
- Consequences of poor connections: reduced efficiency, higher probability of failure, asymmetrical voltage distribution

Energy-related perspective

- Contacting and wiring to the finished stator are considered in parallel
- From an energy point of view, the contacting process is one of the largest consumers in the production of electrical machines

In the context of electromechanical engineering, the process of contacting is usually understood as the electrical and mechanical connection of enameled wires.

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Contacting: Electrical and mechanical connection of enameled conductors

Conductor

- Conductor material
  - Copper (e.g. Cu-ETP)
  - Aluminum
- Area of the cross-section
  - Round wire: 0.0025 – 44.18 mm²
  - Flat wire: 1.5 – 88.8 mm²
- Number of parallel conductors
- Form of the cross-section
  - Round
  - Rectangular
  - profiled

Insulation

- Insulation material
  - Polyesterimide
  - Polyimide
  - Polyurethane
  - Polyvinylacetal
  - Polyetheretherketone
- Temperature index
  - 105 - 240
- Layer thickness
  - Grade 1 - Grade 3
- Overcoat
  - Polyamidimide
  - Polyamide
- „Overcoat enameled wire“
  - Basecoat: Polyesterimide
  - Topcoat: Polyamidimide

Terminal part

- Type of terminal / attachment part
  - Sleeve
  - Cable lugs (Tubular, Compression-, Crimp-)
  - Contact tab
  - Contact pin
  - Bifurcated contact
  - insulation-displacement contact (IDC)
- Base material
  - Copper (e.g. Cu-HCP)
  - Aluminum
  - Nickel
- Coating
  - Tin
  - Nickel

Contact area

- Single wires – Single wires
- Wire bundle – Sleeve (e.g. Tubular cable lug)
- Single wires – Contact element (e.g. Contact tab, Contact pin)
- Compacting (e.g. Strand end)

Electromechanical engineering is confronted with a rapidly increasing number of different contacting tasks.

Typical contacting tasks in the production of electric machines

Contacting of parallel wire bundles

- Variety of enameled round wires, bundles
- Cable lugs
- e.g. pulled-in distributed winding


Graphic: FAPS

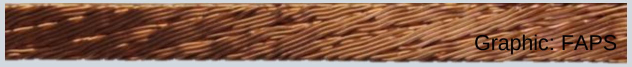
Contacting of high-frequency stranded wires

- Fine, stranded and enameled wires
- e.g. inductive charging technology

Graphic: Rudolf Pack GmbH & Co. KG

Contacting pressed litz wires

- Enameled copper wire bundle
- Pressed into shape (increase filling level)
- e.g. shaped strand windings



Graphic: FAPS

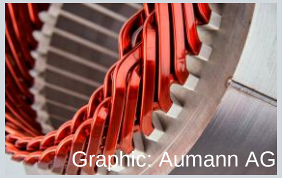
Contacting of single wires

- Enameled round wire
- Enameled flat wire
- e.g. concentrated winding

Graphic: ATOP S.p.A.

Contacting of hairpins

- Enameled flat wires
- High number of contact areas



Graphic: Aumann AG

Material aspects

- Enameled copper conductor
- Enameled aluminum conductor
- Anodised aluminum conductors
- Various insulations
- Mixed compounds (e.g. Al and Cu)

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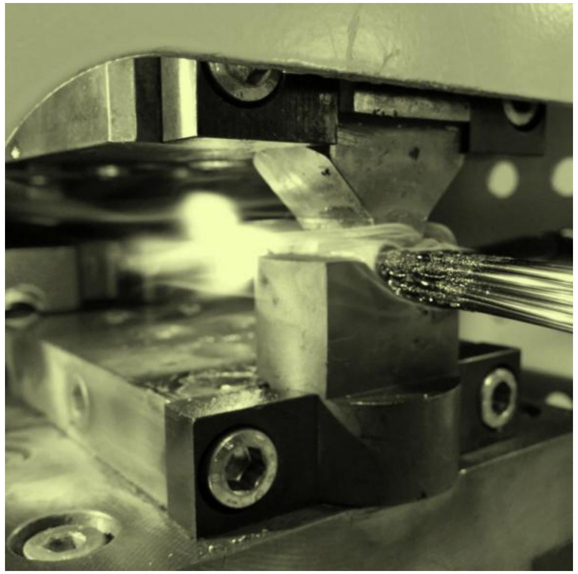

In contacting technology, a distinction can be made between the contact design and the actual contact application.

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Electrical contact

Definition:

An electrical contact is the touching of (two) electrically conductive materials.

Accordingly, two essential tasks result:

- Create a mechanical connection
- The conduction of an electrical current from one surface of a contact to the next

Distinction with regard to contact design

stationary	separable
Examples:	**Examples:**
- Welding	- Switching contacts
- Bonding	- Plugged contacts
- Soldering / Brazing	- Sliding contacts
- Squeezing	
- Adhesive Bonding	

Distinction with regard to contact application

Information technology	Energy technology
Used primarily for the transmission of information, e.g.:	**Used primarily for the transmission of energy, e.g.:**
- PC keyboard	- Motors
- High-frequency plug	- Control units
	- Transformers

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The key requirements for a contact point are low contact resistance and reliable transmission regardless of environmental conditions.

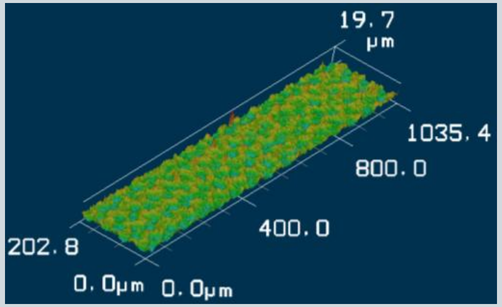
Requirements for the contact point

- Low and constant contact resistance between the contact partners
- Reliable transmission regardless of environmental conditions
- Low heating of the contact points
- Good abrasion resistance of separable contacts
- No sticking and adhesion phenomena with separable contacts (e.g. relays)

Influences on the contact resistance

Shape deviations

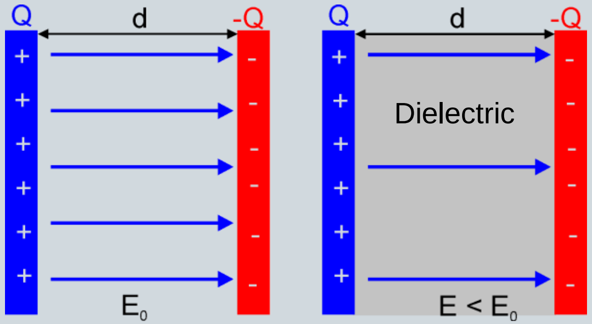
- Form deviation, waviness, roughness
- Current flow only through metallic contact points: a-spots



→ Constriction resistance

Impurity layers

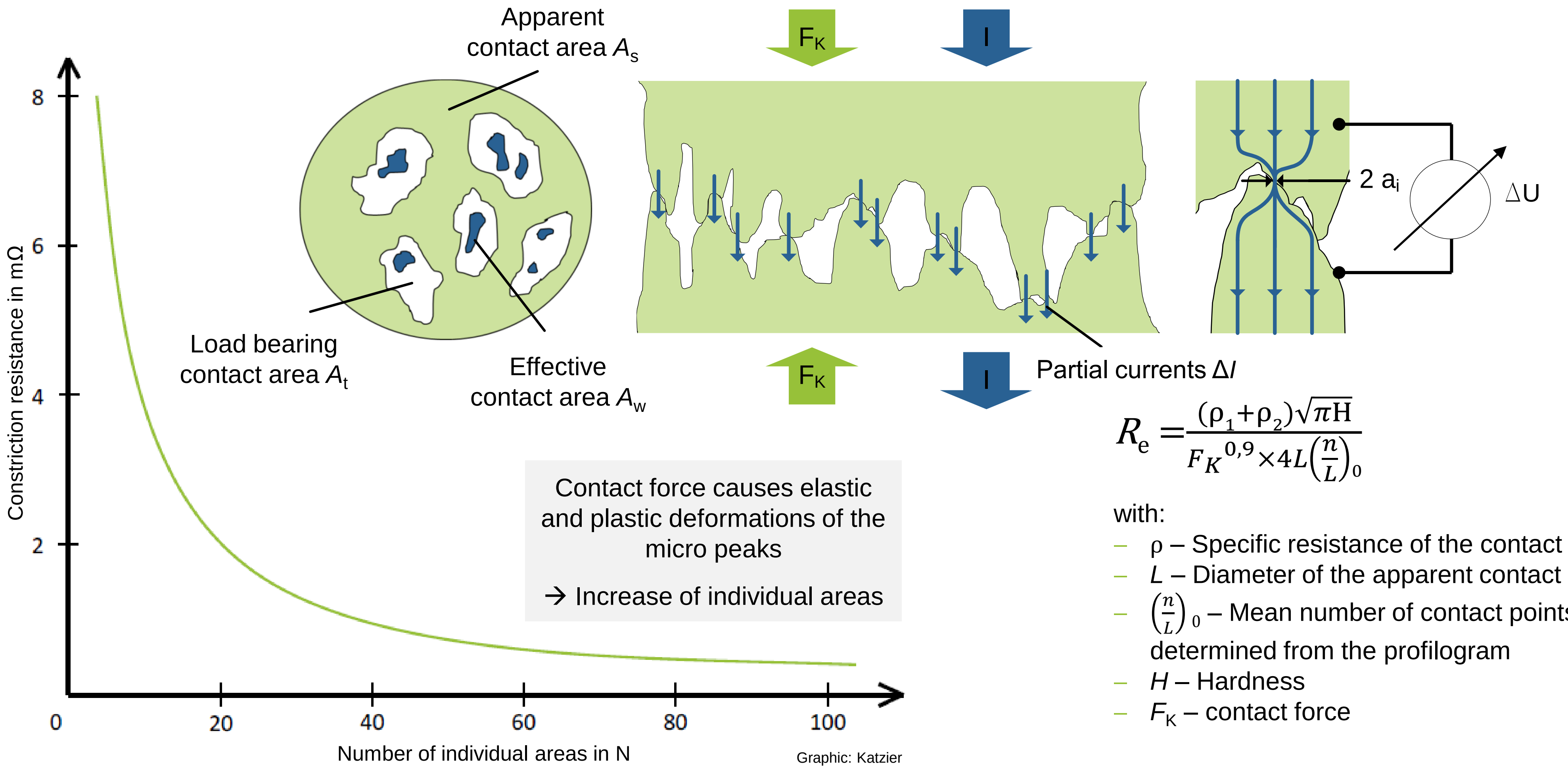
- Presence of impurity layers
- Oxide layers, impurities
- Can be penetrated dielectrically



→ Impurity layer resistance

The contact resistance R_C is mainly influenced by the constriction resistance and the impurity layer resistance.

Contact model (constriction resistance)



Graphic: Katzier

$$R_e = \frac{(\rho_1+\rho_2)\sqrt{\pi H}}{F_K^{0,9} \times 4L\left(\frac{n}{L}\right)_0}$$

with:

- ρ – Specific resistance of the contact partners
- L – Diameter of the apparent contact area
- $\left(\frac{n}{L}\right)_0$ – Mean number of contact points, to be determined from the profilogram
- H – Hardness
- F_K – contact force

Changes in contact resistance can occur for several reasons. One is the electrical breakdown of high-resistance corrosion and contaminant layers, also known as frits.

Contact model (impurity layer resistance)			
A-frits	Generation of a first conductive channel by ■ Mechanical breaking (increase in contact force) ■ Abrasion (increase in relative movements) ■ Dielectric breakdown (increase in electrical voltage)	B-frits	Enlargement of the generated a-frits through further current flow

$$R_F = \frac{d}{A}\rho_F$$

with:
- ρ_F – Specific resistance of the impurity layer
- *d* – Layer thickness
- *A* – Contact area

Determination of R_C by addition of R_e and R_F (for $R_e > R_F$): $R_C = \frac{\Delta U}{I} = R_e + R_F$

The example of contacting with a cable lug shows the various influencing variables on the system of the electrical contact in practice.

Design of the contact elements

- Degree of filling
- Connection mechanism
  - Positive fit
  - Force fit
  - Material locking
- Shape deviations

Conductors to be connected

- Geometry
- Insulation

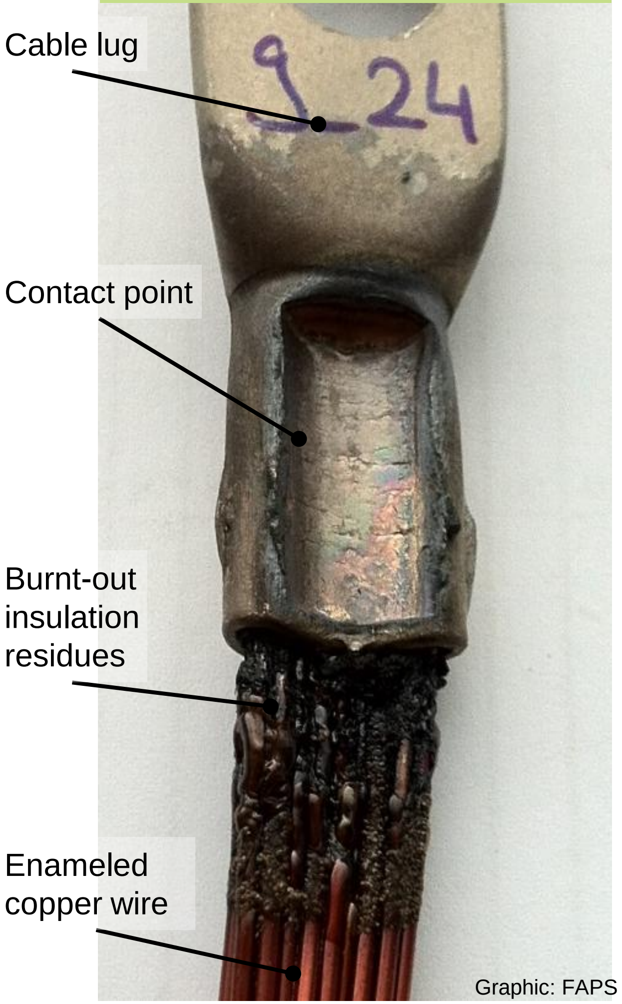
Contacting process

- Contact force
- Contact length
- Removal of impurity layers
- Formation of impurity layers

Electrical load

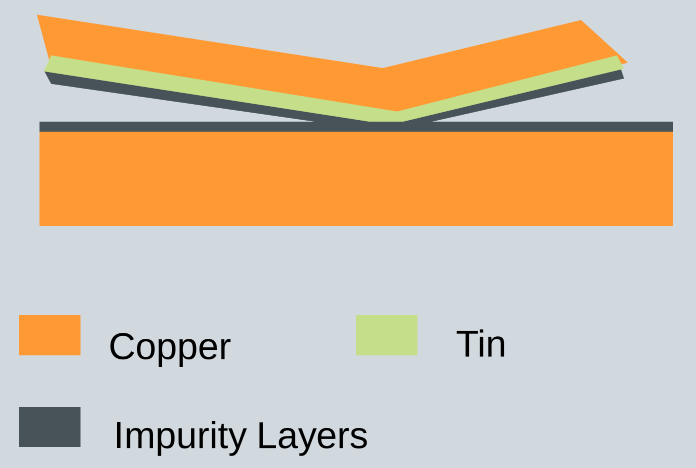
- Current level
- Current duration

Contacted cable lug



Graphic: FAPS

Contact layers



Environment

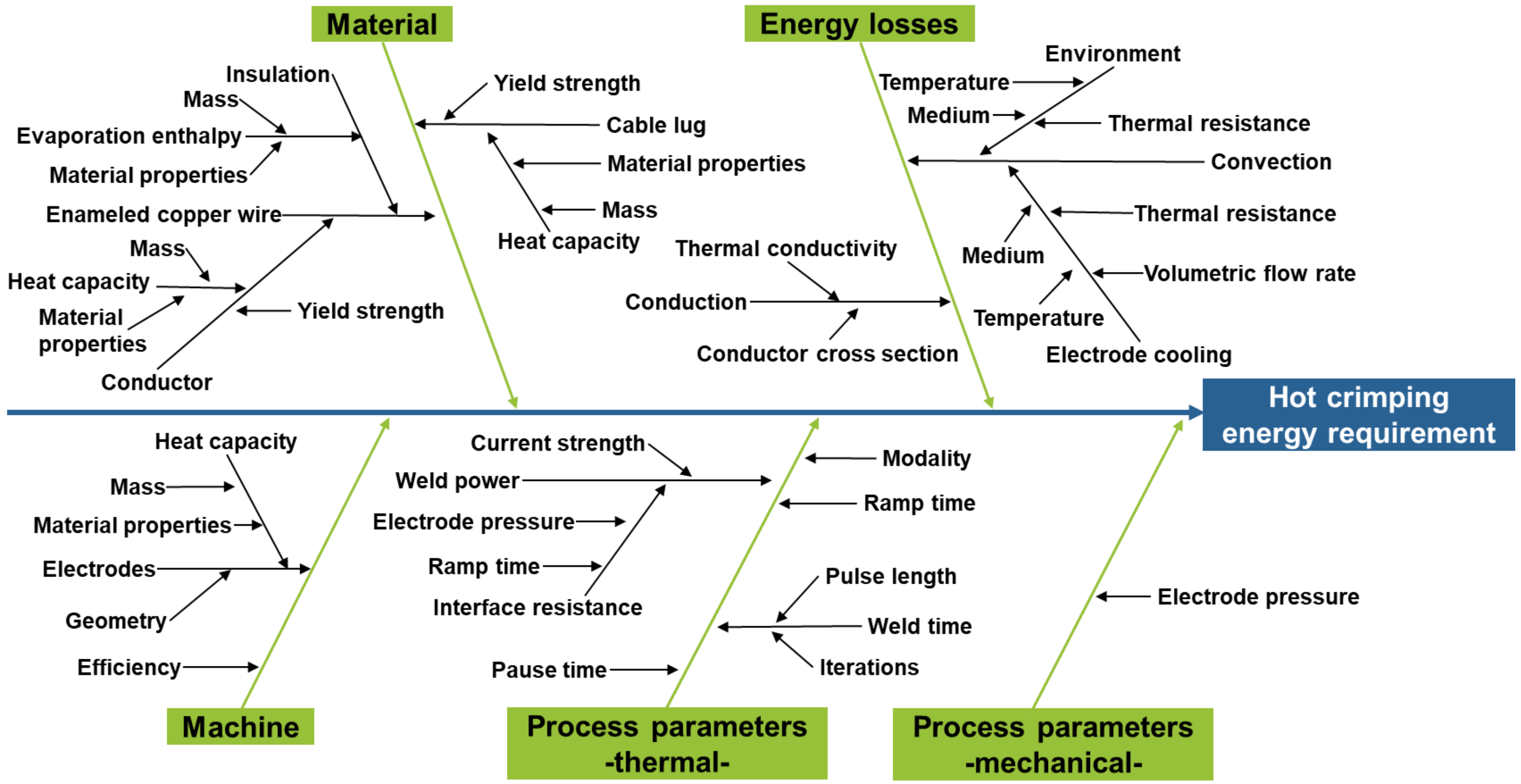
- Temperature
- Humidity
- Vibration
- Corrosion

Mechanical load

- Static
- Dynamic

The example of hot crimping shows the enormous complexity of the contacting process and the large number of influencing factors.

Hot crimping as an example of the contacting process complexity



- Identification of the most important factors based on theoretical considerations prior to testing
- Considerations for welding power, weld time, number of iterations, electrode pressure, pause time
- Considerations for interactions between influencing factors
- Possibility of non-linear relationships

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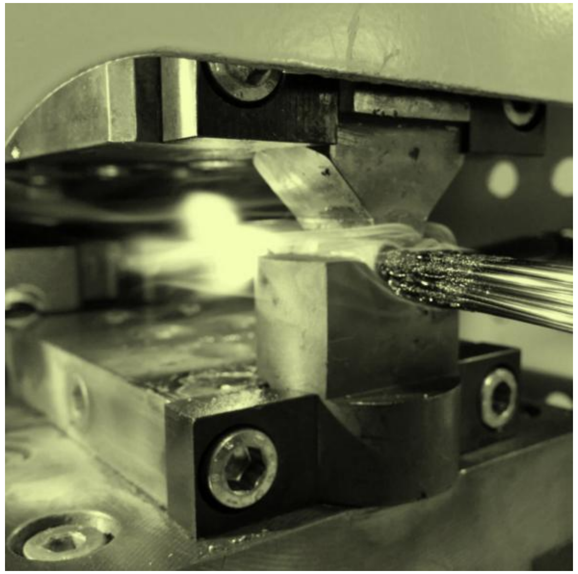
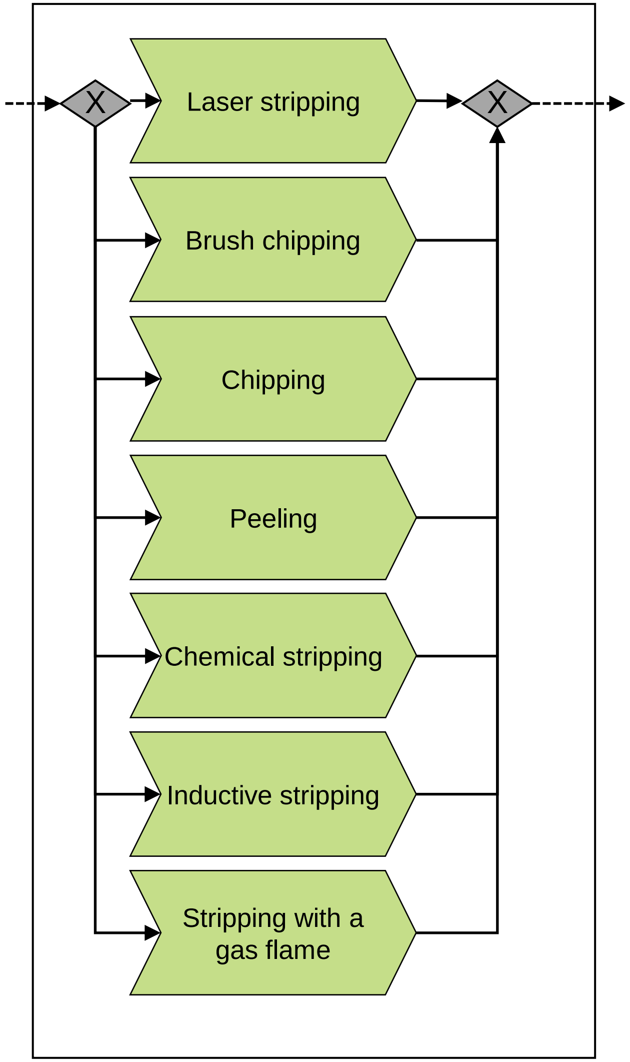
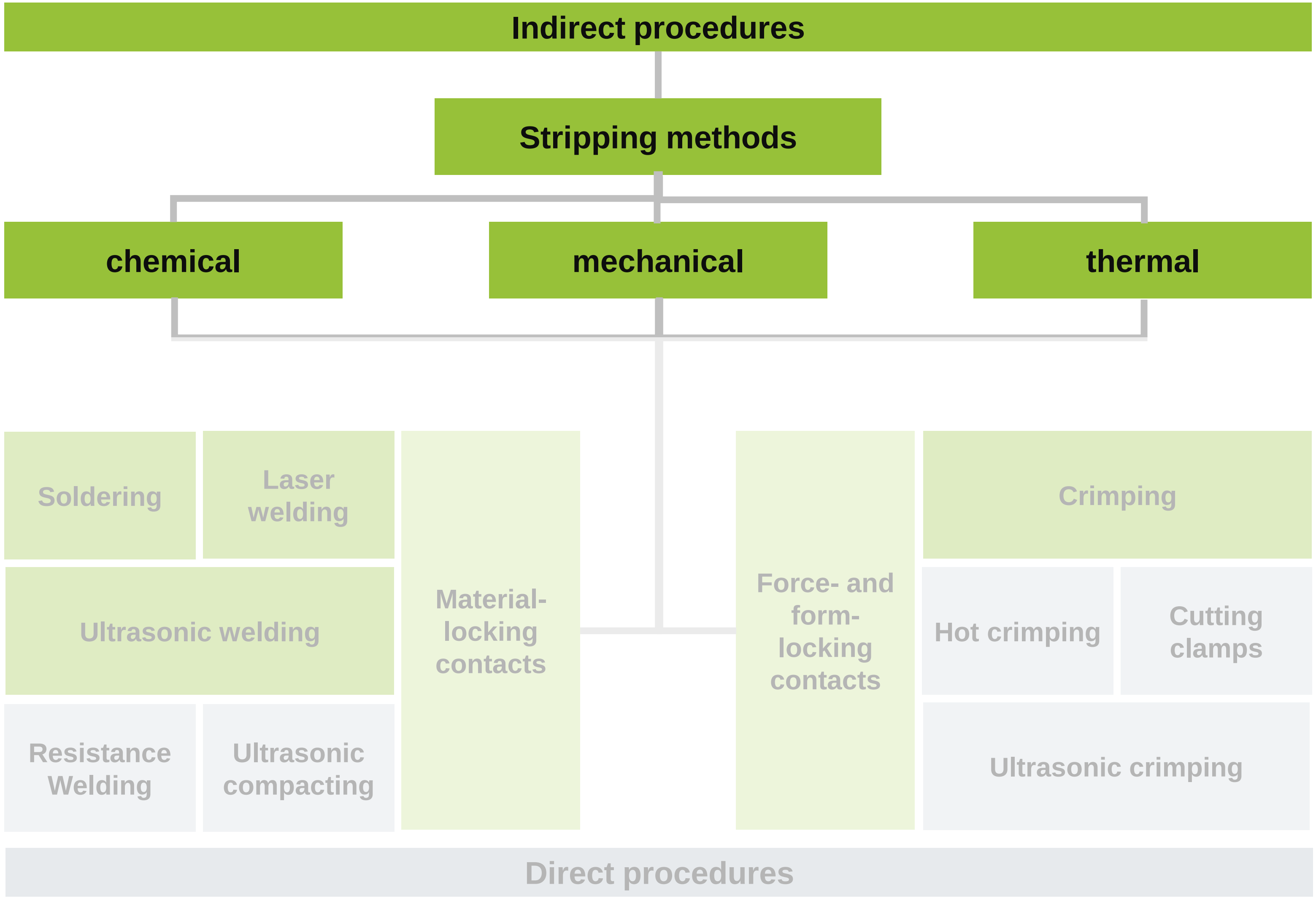


In indirect contacting processes, the enameled copper wires are first stripped, whereby the stripping methods can be divided into three main groups.



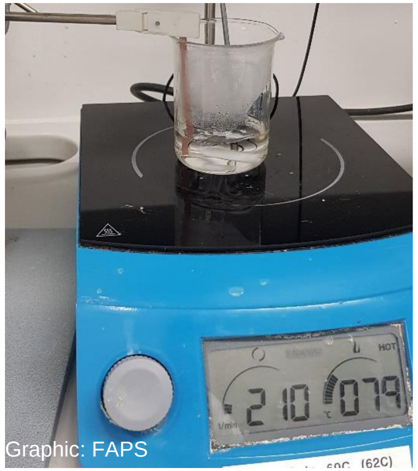


The insulating layer can be removed from the enameled copper wire in different ways.

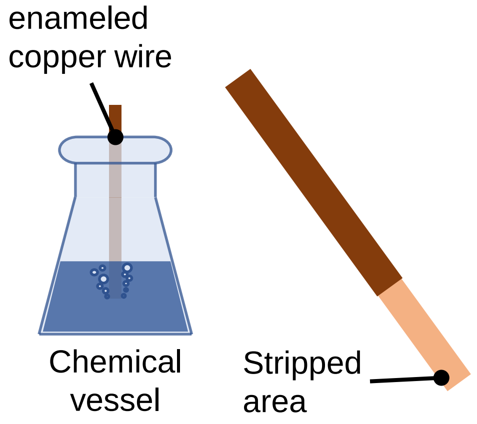
Chemical stripping

Insulated conductor ends are exposed to the chemical "stripper" for a defined exposure time and then cleaned

- Chemical strippers have a strong dissolving power for Novolak- and epoxy-based enamels and polyimides
- At higher application temperatures of approx. 75 °C, also strongly cross-linked enamels
- Consist mainly of solvents (e.g. dimethyl sulphoxide) and bases (tetramethylammonium hydroxide)



Graphic: FAPS



Advantages

- Comparatively cost-effective and low effort
- Application in prototype construction
- Reproducible results

Disadvantages

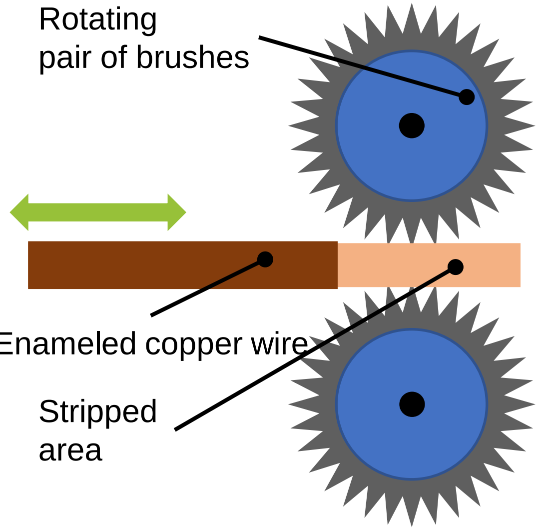
- Very long process times (several hours depending on the degree of insulation and material)
- Handling of substances that are harmful to health
- Heating the "stripper" to temperatures almost below the flash point
- Not suitable for wire bundles, stranded wires or thermoplastic insulation (e.g. PEEK)

The insulating layer can be removed from the enameled copper wire in different ways.

Rotating brushes

The enameled copper wire is inserted between two opposing rotating steel brushes, whereby the insulation layer is removed and vacuumed off.

- Rotating steel brushes are arranged opposite each other
- Distance is adapted to the wire diameter
- Inserting the insulated copper wire
- Removing the insulation due to friction



Advantages

- Low costs
- Simple process control
- Easy to use
- Suitable for various cross-sections
- Applicable for small quantities
- State of the art

Disadvantages

- Fluctuating removal quality and tool wear
- Damage to the wire due to mechanical forces
- Danger of insulation residues
- Undefined transition between stripped and insulated area
- Damage to the conductor surface by brush wires
- Not suitable for melting thermoplastics

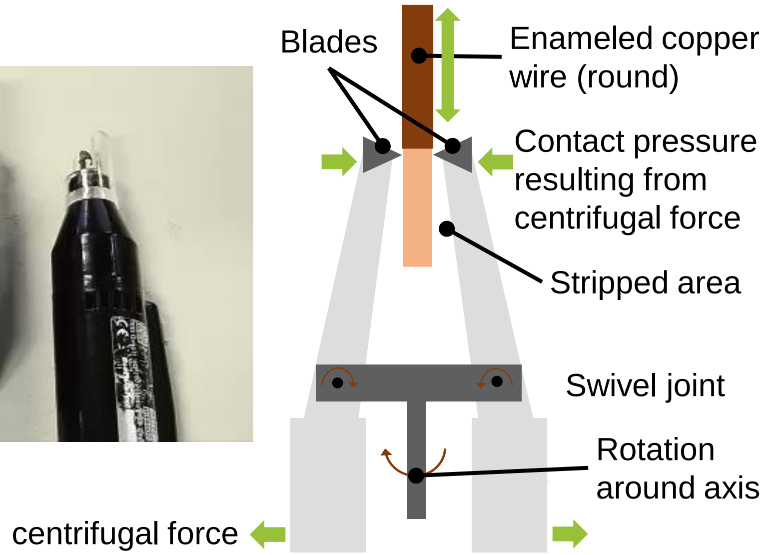
The insulating layer can be removed from the enameled copper wire in different ways.

Rotating blades

The round enameled copper wire is inserted between two rotating blades.

- Closing the blades by rotation
- Varying the rotation speed leads to varying contact pressure
- Suitable for various cross-sections, but mainly round wires due to kinematics

Video: FAPS



Advantages

- Easy handling
- Applicable for small quantities
- Suitable for a wide range of insulation types
- Defined transition area

Disadvantages

- Tool wear
- Not suitable for flat wires
- Damage to the wire surface due to penetration of the blades into the copper material

The insulating layer can be removed from the enameled copper wire in different ways.

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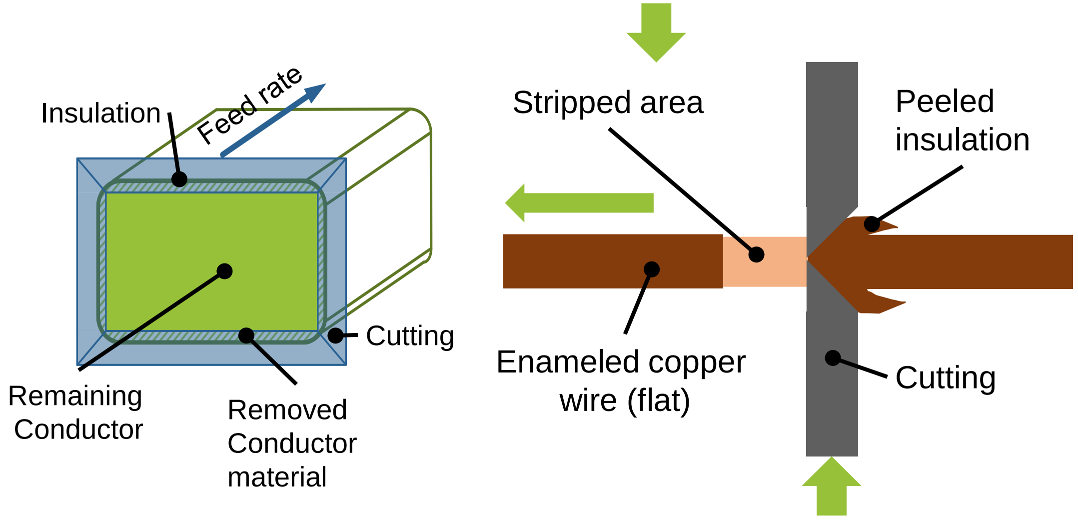
Peeling

Blades cut into the copper material in a rectangular geometry and realize a feed movement along the conductor, whereby the insulation is removed without leaving any residue.

- Kinematic structure usually consisting of four cutting edges
- Cutting edges penetrate the insulation and cut into the conductor material, linear feed movement of the cutting edges along the conductor or of the conductor along the cutting edges leads to peeling of the insulation
- Due to the shape of the flat copper conductors (radii at edges), removal of conductor material cannot be avoided



Advantages

- Easy to use
- Defined cutting line
- Defined transition area
- Simple process principle
- Independent of insulation type
- residue-free removal

Disadvantages

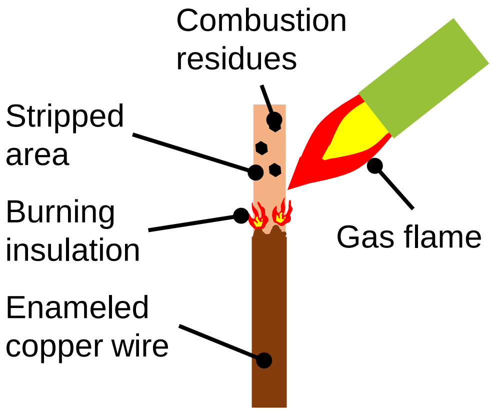
- Removal of conductor material by penetration of the cutting edges into the copper material
- Depth must be adapted to the edge radius
- Tool wear
- Challenging positioning of the cutting edges

The insulating layer can be removed from the enameled copper wire in different ways.

Thermal stripping using a flame

The insulation is burnt off using a soldering torch

- Necessary temperature depends on the insulation used (approx. 450 °C)
- State of the art in many areas, especially when using the brazing process for contacting



Advantages

- Easy handling
- No damage to the copper material
- No additional tools required for brazing processes

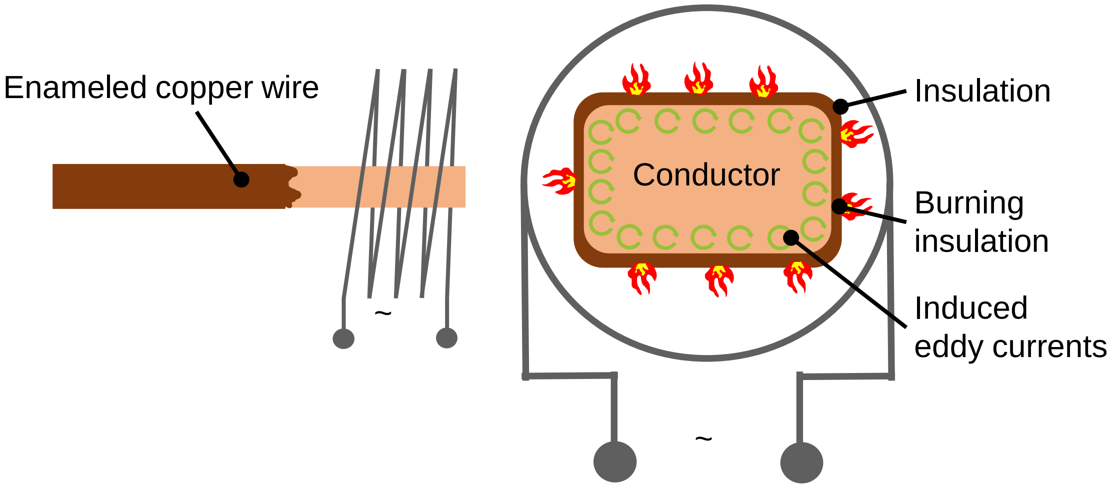
Disadvantages

- High amount of combustion residues
- Strong development of toxic vapours/gases
- Undefined transition area between stripped and insulated area
- Cleaning of wires necessary

The insulating layer can be removed from the enameled copper wire in different ways.

Inductive heating

The conductor is placed in the center of a coil that is connected to a high-frequency alternating voltage. The resulting alternating magnetic field induces eddy currents. The resulting high current density on the surface leads to severe heating of the conductor and burns the insulation.



Advantages

- Short process times
- Non-contact method

Disadvantages

- Oxidation surface
- Heat input damages insulation outside the area to be removed
- Combustion residues

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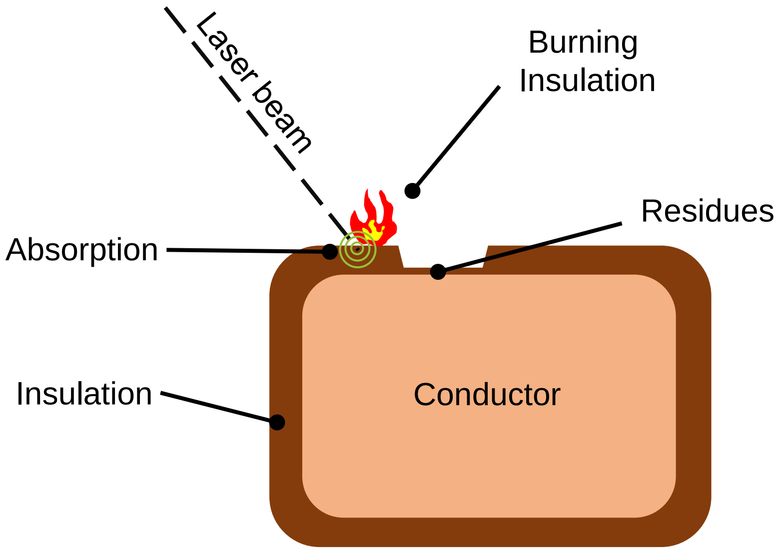
The insulating layer can be removed from the enameled copper wire in different ways.

Laser: Long-wave infrared radiation

Long-wave infrared radiation is reflected by the copper conductor, while it is absorbed in insolation material. This leads to burning of the insulation.

- Use of CO_2 lasers (λ ~ 10,600 nm, pulsed)
- Burning the insulation material
- No interaction with the conductor
- Insulation residues in the size range of the wavelength



Advantages

- Non-contact method
- High removal rate
- Defined transition area

Disadvantages

- Insulation residues in the area of λ/4
- Complex process
- High investment costs

The insulating layer can be removed from the enameled copper wire in different ways.

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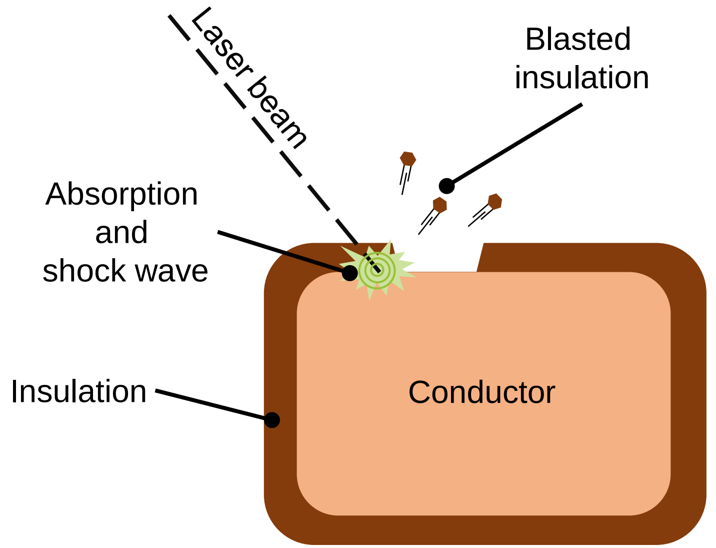
Laser: Near infrared radiation

Near infrared radiation is partially absorbed by the surface of the copper conductor after passing through the insulation. The pulsed laser power causes the copper material to heat up abruptly and expand without melting.

- Use of solid-state lasers (e.g. Nd:YAG laser, λ ~ 1,064 nm) in pulsed operation
- Shock waves are generated which cause the insulation to flake off



Advantages	Disadvantages
Non-contact method	Complex process
Free of residues	High investment costs
Defined cutting line	
High removal rate	

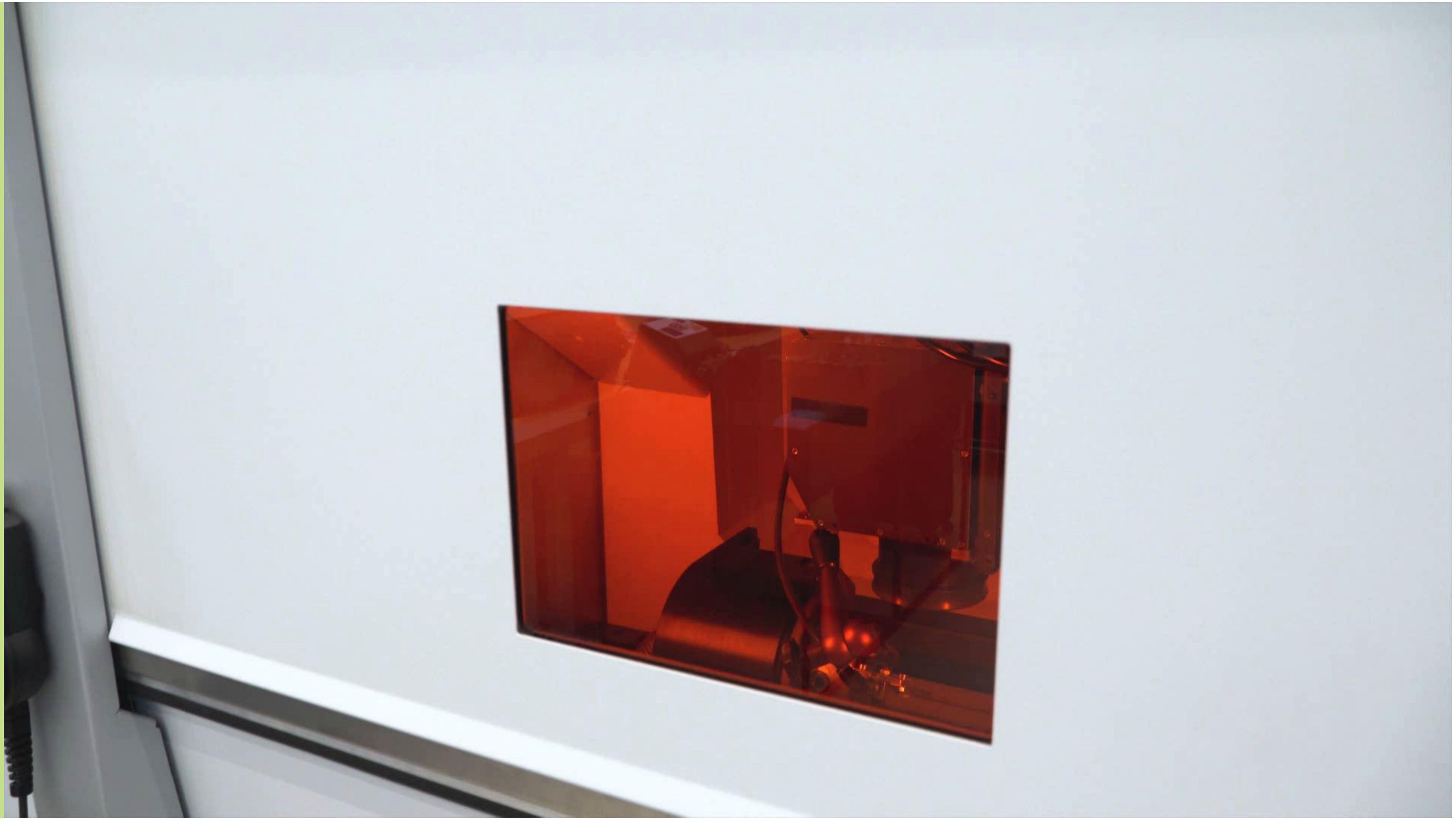
Short-pulse lasers in the infrared wavelength range can be used for the removal of the insulating coating.

Process

Removal of the paint insulation with a short-pulse laser

Video: Trumpf

youtube.com/watch?v=kW0zPOvUY2M



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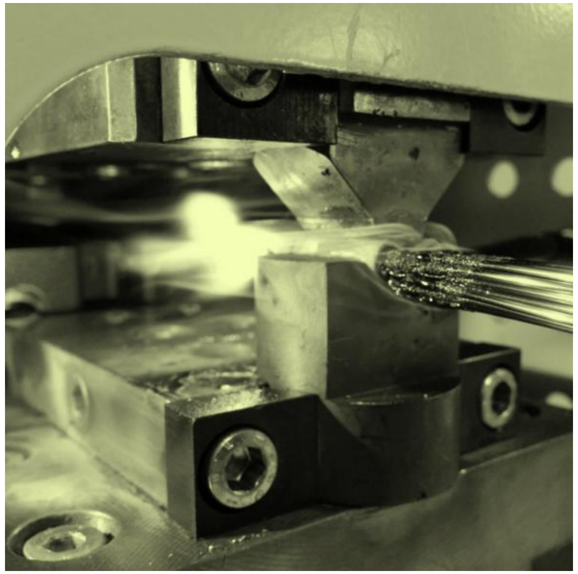

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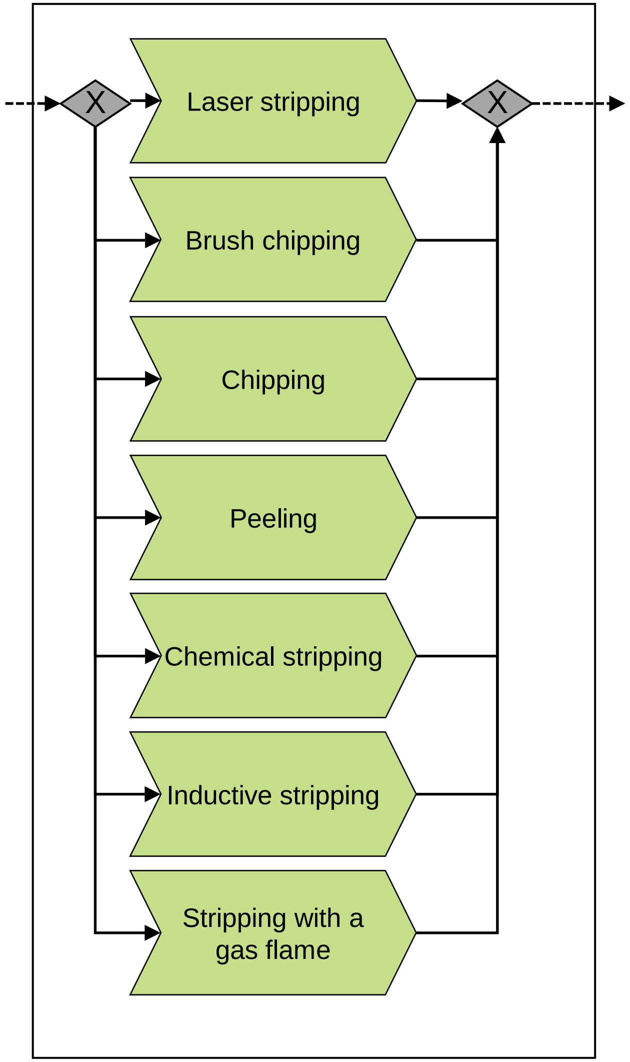
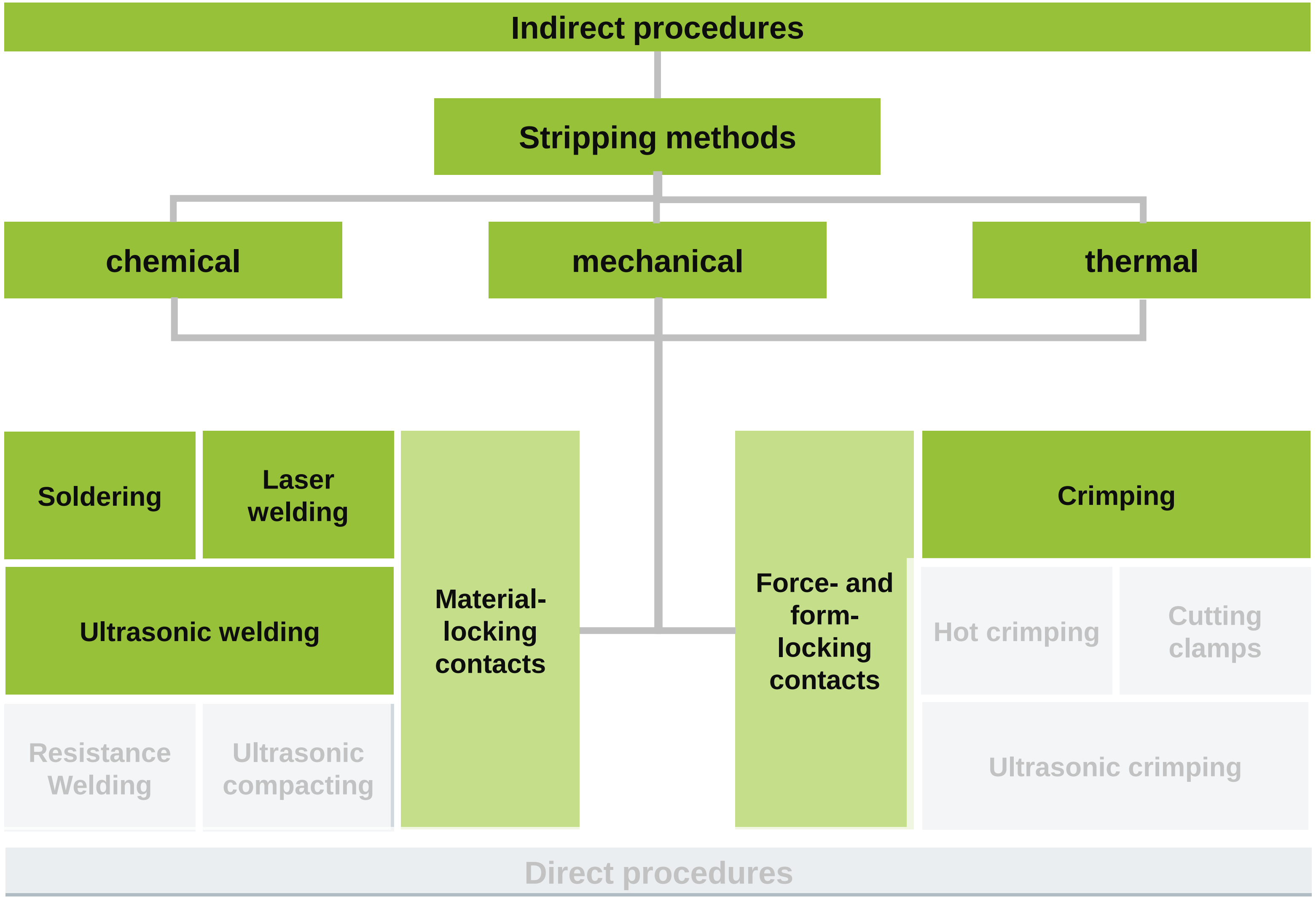


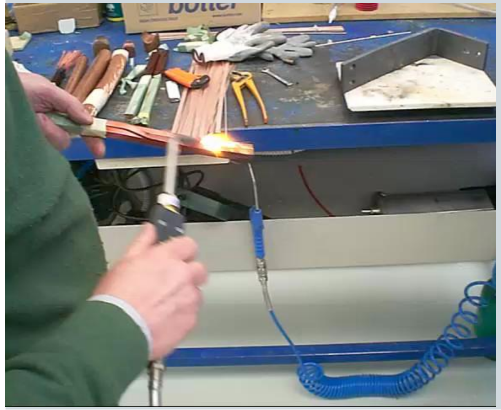


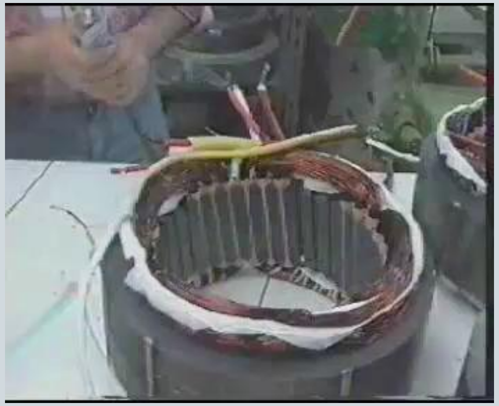
Flame soldering is a widely used contact method in the production of electrical machines, especially for high variety or small quantities.

(Flame) brazing

- Soldering can be divided into soft soldering and hard brazing
  - Soft soldering: Joining process using solders with a liquidus temperature of 450°C or less.
  - Hard Brazing: A joining process using brazing alloys with liquidus temperatures above 450°C.
- Flame brazing and, in some cases, solder bath brazing are primarily used for contacting in the production of electrical machines
- Thermal paint stripping with a flame is usually carried out in advance
- Solders e.g. CuSn, AlSi

Videos: Oxyweld





Advantages

- Joining of different materials
- Lower heat input than welding > Less distortion
- Reliable process

Disadvantages

- Vibrations and thermal load changes can lead to cable breakage
- Risk of cold junction with larger cross sections
  - High contact resistance
  - Energy loss and contact heating (fire hazard)
- Long process times
- Lower strength compared to welding

The laser welding process is mainly used for molded coil windings such as hairpins.

Laser beam welding

- Use of high-brilliance solid-state lasers (e.g. disc lasers) in the infrared wavelength range
- Use of high welding power to form a keyhole despite high reflection of the laser radiation at the start of the process
- Melting of the contact partners and formation of a material locking bond
- Mainly used for flat wires and single wires
- Further optimization of the laser beam welding process through the use of green / blue lasers

Videos: Keyence

Advantages

- Extremely short process times
- Scanner-based beam guidance
- High process stability
- Realization of high welding depths
- Contactless process

Disadvantages

- High system investment
- Sensitive to insulation residues and positioning errors

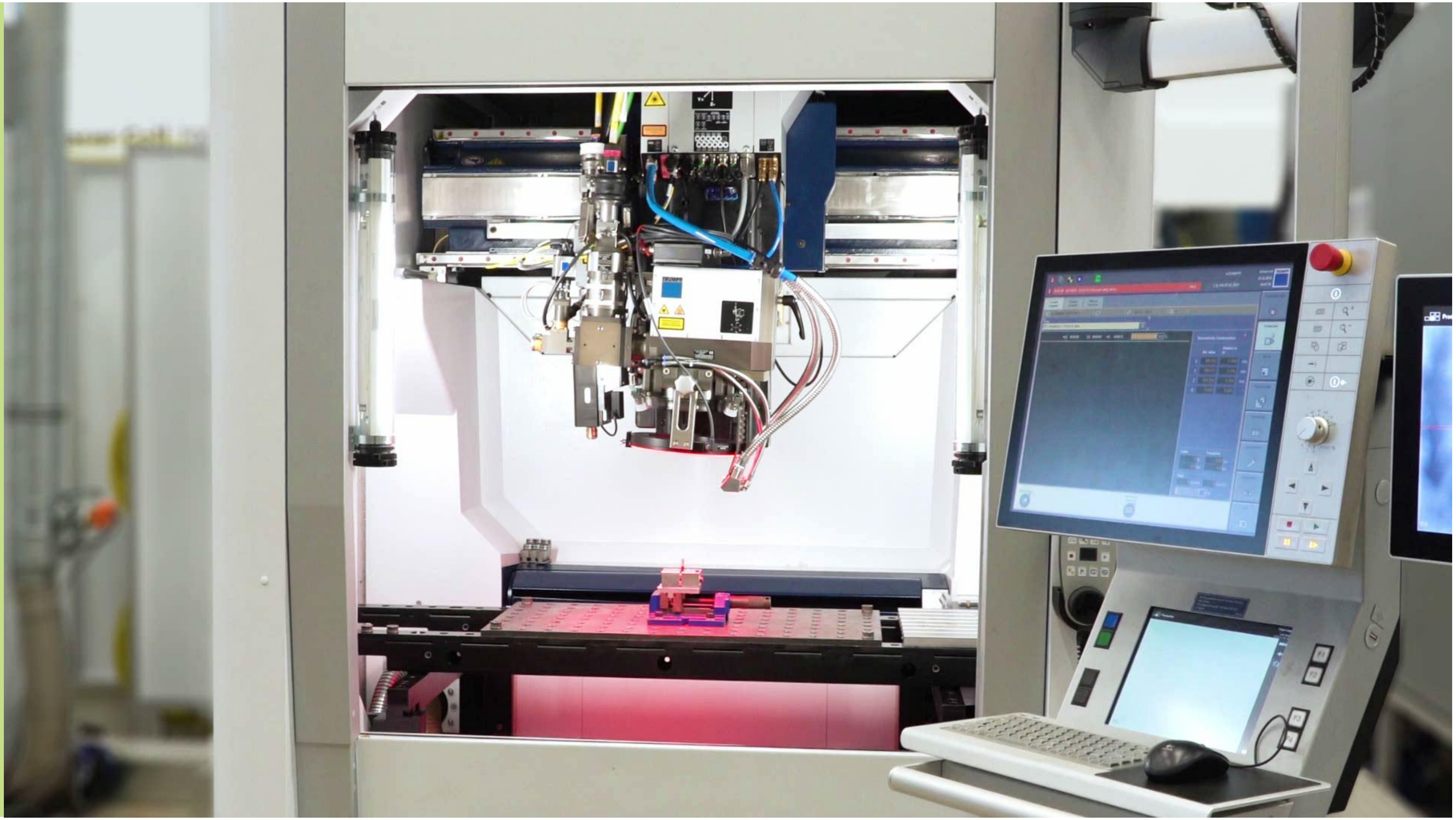
Laser beam welding of preformed coil windings shows potential for producing the large number of contact points on preformed coil windings.

Process

Contacting of flat conductor-based preformed coil windings using the laser welding process

Video:
Trumpf

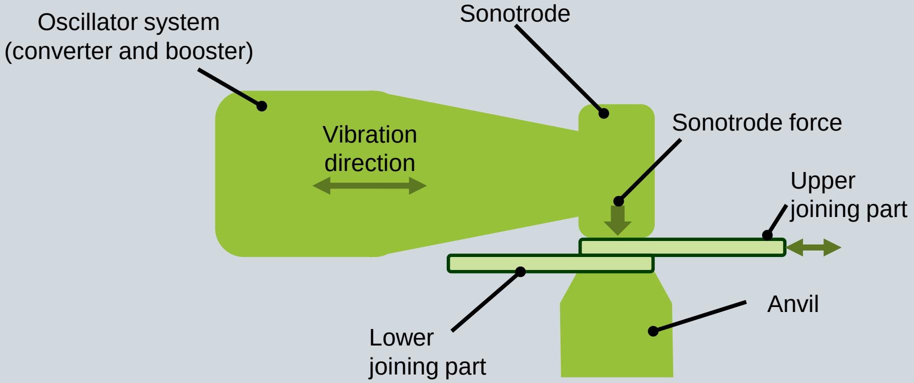
youtube.com/watch?v=kW0zPOvUY2M



Classic ultrasonic welding of metallic materials is only suitable for non-insulated conductor materials.

Ultrasonic welding	
	Advantages

- Cold welding through the effect of high-frequency mechanical vibration energy and simultaneous pressure
- Heating of the workpieces below the melting points of the joining partners
- In ultrasonic welding, metal joining partners are positioned between a fixed anvil and the high-frequency vibrating sonotrode, which simultaneously applies the required welding pressure
- The welding process is divided into three stages: elastic deformation, plastic deformation and oxide layer removal as well as the formation of the welding zone



Advantages

- Contacting of different materials (e.g. Cu + Al)
- Good automation potential
- Very good mechanical and electrical connection quality
- Good process control

Disadvantages

- Stripping of enameled wires necessary (smaller round wires, however, also without stripping)
- Poor accessibility due to tool design
- Vibrations introduced can damage joining partners

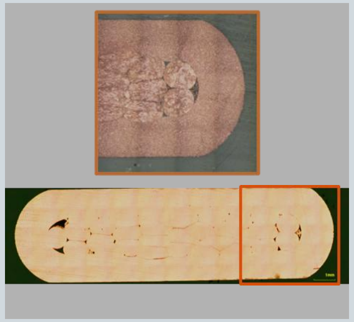
FAPS

In the mechanical crimping process, the connection is formed by the plastic deformation of the crimp sleeve that is used.

(Cold) crimping

- Force-fit connection between the cable conductor and the crimp connection elements of the connector or the connecting element
- Connection created by plastic forming of the crimp sleeve
- Connection types: both stripped conductor wires and litz wires can be used
- Bundled stranded wires are preferably used
- The crimping process is carried out using special crimping tools
- Force- and path-controlled process monitoring

Graphic: Klaus Faber AG, Gustav Klauke GmbH, FAPS



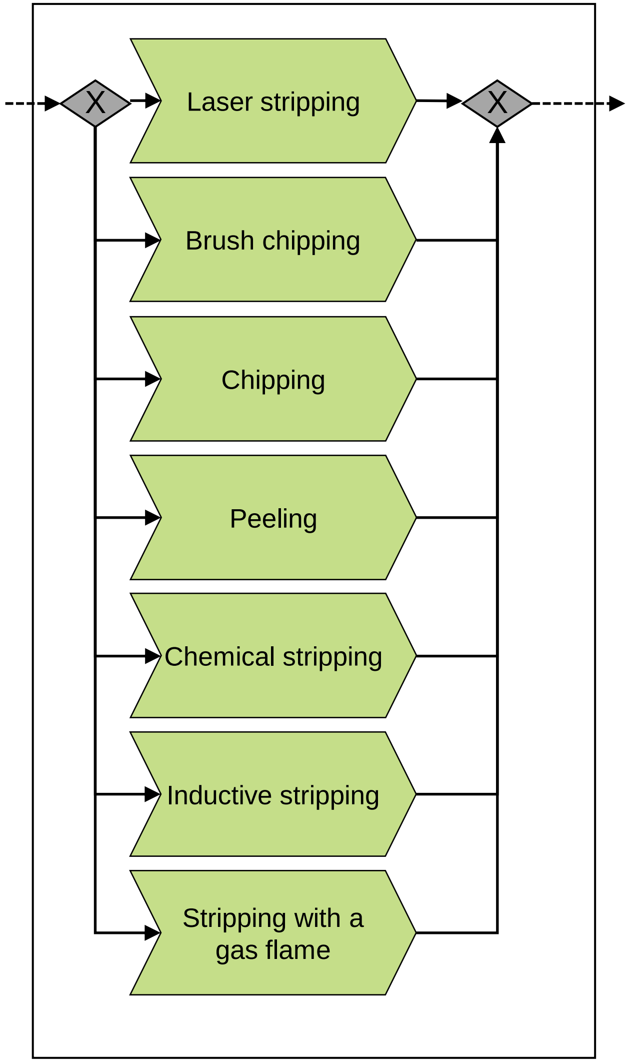
Advantages

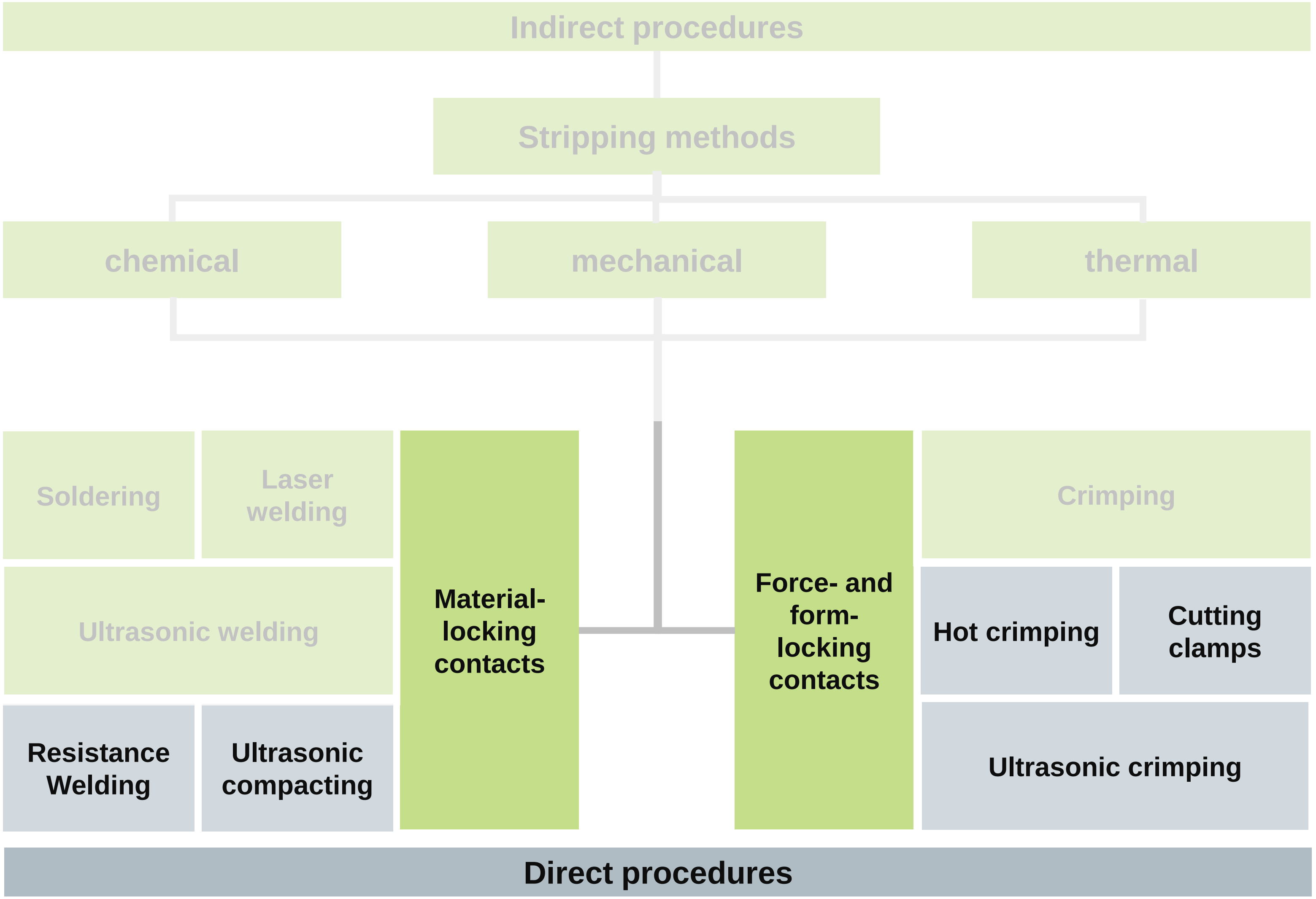
- Contacting of different materials (e.g. Cu + Al)
- Good automation potential
- Cost-effective production
- Good process control

Disadvantages

- Capillary formation
  - Oxidation
  - Increased contact resistance
- Stripping of enameled wires necessary
- Vibrations may cause the cable to break

In indirect contacting processes, the enameled copper wires are first stripped, whereby the stripping methods can be divided into three main groups.



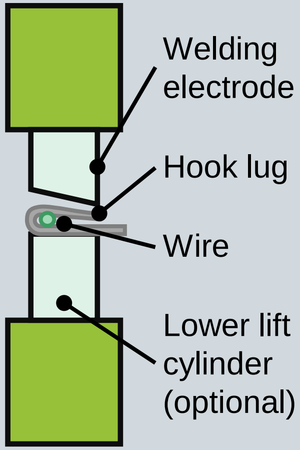


The resistance welding process is particularly suitable for contacting round individual wires.

Resistance welding

Joule heating of the electrodes and workpiece by pulsed current supply

- Direct current and alternating current possible
- Multi-stage process (several pulses)
  - Thermal stripping
  - Welding
- Adjusting the electrodes



Video: AMADA

https://www.youtube.com/watch?v=0LC_getmplg

Advantages

- Contacting of enameled wires (round / flat) possible by shunt connection via hook lug
- Good automation and process control

Disadvantages

- Mostly only suitable for single wires
- Special connection geometries necessary
- Principle-based cross-sectional constriction

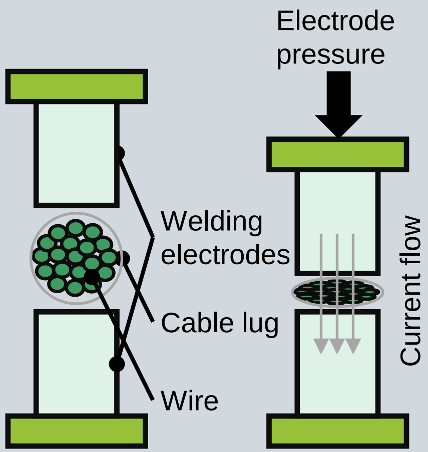
The hot crimping process reflects the current industrial standard for contacting enameled wire bundles.

FAPS

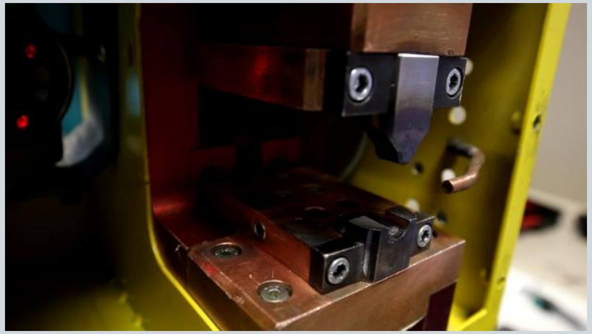
Hot crimping

Hot crimping is a type of resistance welding in which a conductive connection part (a cable lug, terminal, or a simple round sleeve) is used.

- The heat generated by the flowing current (Joule heating) causes the insulation to evaporate and the wires to bond together
- By integrating the process step of stripping, productivity in the production of electrical connections is increased



Video: FAPS



Advantages

- Stripping and contacting of enameled wire bundles
- Good automation and process control
- Good mechanical and electrical connection quality
- Short process times

Disadvantages

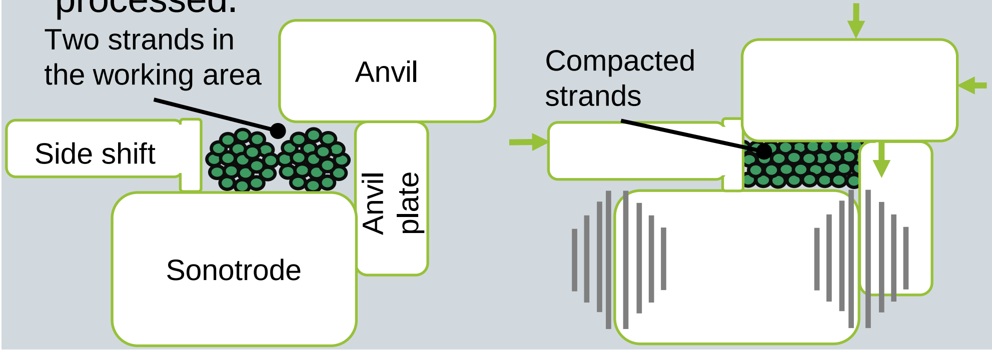
- High tool wear (depending on electrode material)
- Combustion residues from the insulation
- High process energy requirement

Ultrasonic compacting / splicing can be used both in the production of wiring systems and in contacting technology for electric drives.

FAPS

Ultrasonic compacting / splicing

- This is based on the basic principle of ultrasonic welding of metals. A defined lump geometry is produced by the kinematics, consisting of sideshift, anvil, sonotrode and anvil plate. and anvil plate. This results in plastic deformation of the strands below the melting temperature.
- Genuine metallic connection, without any foreign material, directly connected to each other in a gas-tight manner
- Ultrasonic splicing → Two non-insulated strands are contacted with each other (spliced)
- Ultrasonic compacting → A stranded wire is "compacted" and, if geometrically possible, stripped in the process due to the process heat. The compacted strand can be further processed.



Advantages

- Contacting of different materials (e.g. Cu & Al)
- Good automation capability and process control
- Very good mechanical and electrical connection quality
- Stripping in the process possible

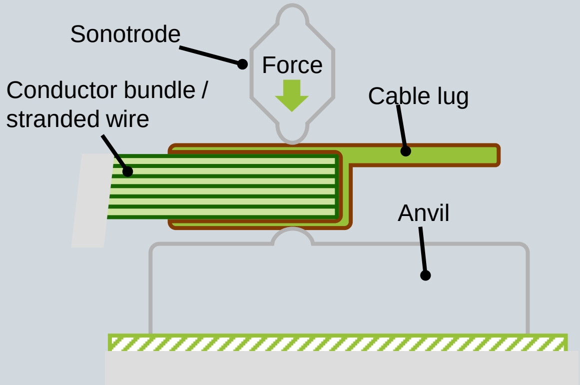
Disadvantages

- Strong damping effects when using insulated strands
- Poor accessibility due to tool design
- Vibrations introduced can damage joining partners

Ultrasonic crimping is an energy-efficient alternative to the hot crimping process.

Ultrasonic crimping

- Combination of ultrasonic welding with mechanical crimping
- Use of a connecting part (cable lug, sleeve, ...)
- The applied force of the sonotrode causes a mechanical deformation of the joining part in the z-direction and a cold welding of the joint.
- High-frequency mechanical vibrations of the sonotrode in the range of 20 to 35 kHz simultaneously leads to friction between the connecting part and the enameled copper wires
- Generated thermal energy leads to melting and burning of the insulating layer



Video: FAPS

Advantages

- Stripping and contacting of enameled wire bundles
- Low process energy requirement
- Short process times
- Low tool wear

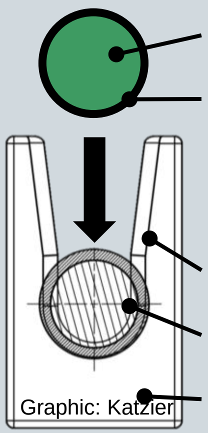
Disadvantages

- Burning residues of the insulation
- Fatigue failure of cable lugs is possible due to vibration coupling and damping effects

The insulation displacement method is almost never used in the production of electrical machines due to its susceptibility to vibration.

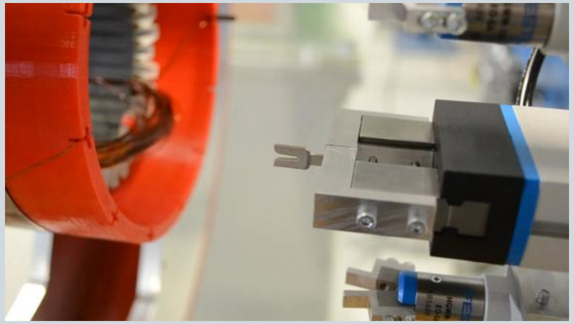
Insulation displacement method

- Mechanical stripping of the enameled wires during the insertion process on the sharp-edged legs of the insulation displacement connectors
- Contact is made by applying pressure (force-fit) to insulation displacement connector
- Insulation displacement connectors have mechanical pretension
- Insulation displacement connectors must be matched to the wire cross-section



Graphic: Katzier

Video: FAPS



Advantages

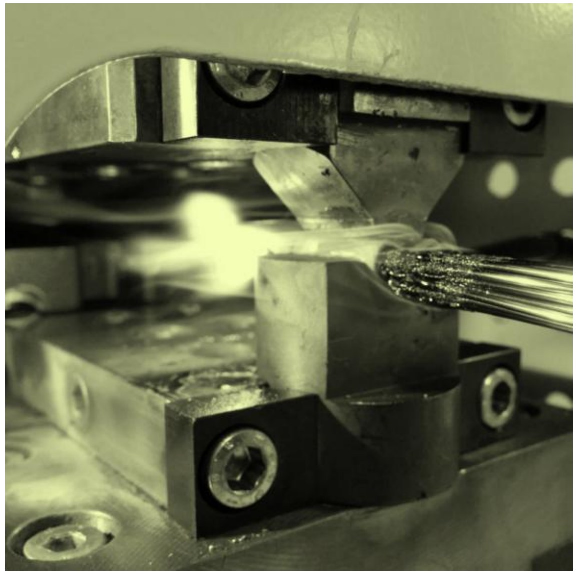
- Possibility of full automation
- Low process costs

Disadvantages

- No multi-wire connections
  (max. two wires of the same diameter)
- Lack of process reliability for the automotive sector
- No strain relief
- Sensitive to vibration

This lecture provides an overview of the physical principles of electrical contact and the most common contact technologies used in electromechanical engineering.

- Contacting in the production of electric motors and machines
- Basics of electrical contacts
- Insulation stripping process technologies
- Process technologies for the production of electrical contacts
- **Testing of electrical contacts in electromechanical engineering**
- Research activities in the field of contacting



There are no explicit standards for testing contact points in electromechanical engineering, which is why testing is based on various standards.

Reference	Standard	Description
General (Plug connectors)	DIN EN 60512-1	Plug connectors for electronic equipment: Measuring and test methods: General
General (Crimp connections)	DIN EN 60352-2	Crimp connections - General requirements, test methods and application notes
General (Welded joints)	DIN EN ISO 17635-08	Non-destructive testing of welded joints - General rules for metallic materials
Mechanical strength (Crimp connections)	DIN EN 60512-16-4	Mechanical tests on contacts and connections: Tensile strength of crimp connections
Mechanical strength (Weld seam)	DIN EN ISO 4136-02	Destructive testing of welded joints on metallic materials: transverse tensile test
Contact resistance (Plug connector)	DIN EN 60512-2-1	Connectors for electronic equipment - Measurement and test methods: Electrical continuity and contact resistance tests
Long-term stability (General)	DIN EN 60068-1	Environmental influences: General information and guidelines
Corrosion test	DIN EN 60068-2-52	Environmental testing: Test method, test Kb: Salt spray, cyclic (sodium chloride solution)
Climate impacts	DIN EN 60512-11-1	Electromechanical components for electronic equipment - Measurement and test methods - Part 11: Climatic tests; Section 1: Test 11a: Climate impacts

The qualification of contact points can be carried out mechanically, electrically and with regard to long-term stability.

Static mechanical testing

Pull-out tests

Dynamic mechanical testing

- Temperature shock test
- Vibration test

Corrosion testing

Salt spray test

There are several challenges in the qualification of contact points:

- Non-destructive testing
- Difficult to measure resistances
- Overfulfilment of standards
- Minor changes

Graphics & Videos: FAPS

Imaging methods

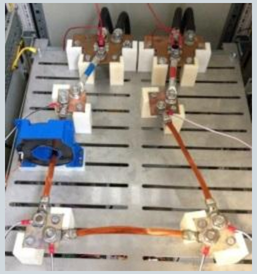
- Microsections
- X-ray CT

Dynamic electrical testing

- Slow-Motion -Bending Test
- Online monitored dynamic resistance test

Static electrical testing

- Measurement of contact resistances at high currents
- Nanoohmmeter



In future, even minor differences in quality are to be detected through targeted pre-damage of contacting samples.

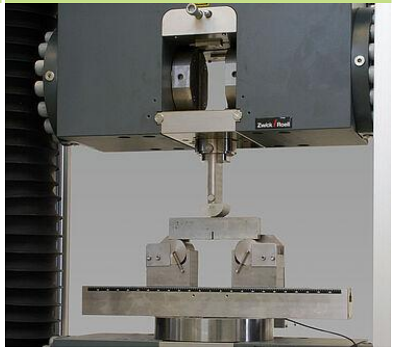
Example: Pre-damage of test specimens

Formation of harmful foreign layers / increase in contact resistance, e.g. due to

Corrosion (salt spray test)	Temperature change	Mechanical stresses

Graphic: FAPS

Graphic: CTS Umweltsimulationen



Graphic: ZwickRoell GmbH & Co. KG

Online monitored dynamic resistance test

- Cyclical heating of the crimps using Joule heat
- Online measurement of resistances/ temperatures
- 1000 cycles
- T_{peak} = 120 °C
- T_{basis} = 35 °C

Static measurement of resistances

Measurement of the contact resistances

- Before,
- Between and
- After carrying out environmental simulations

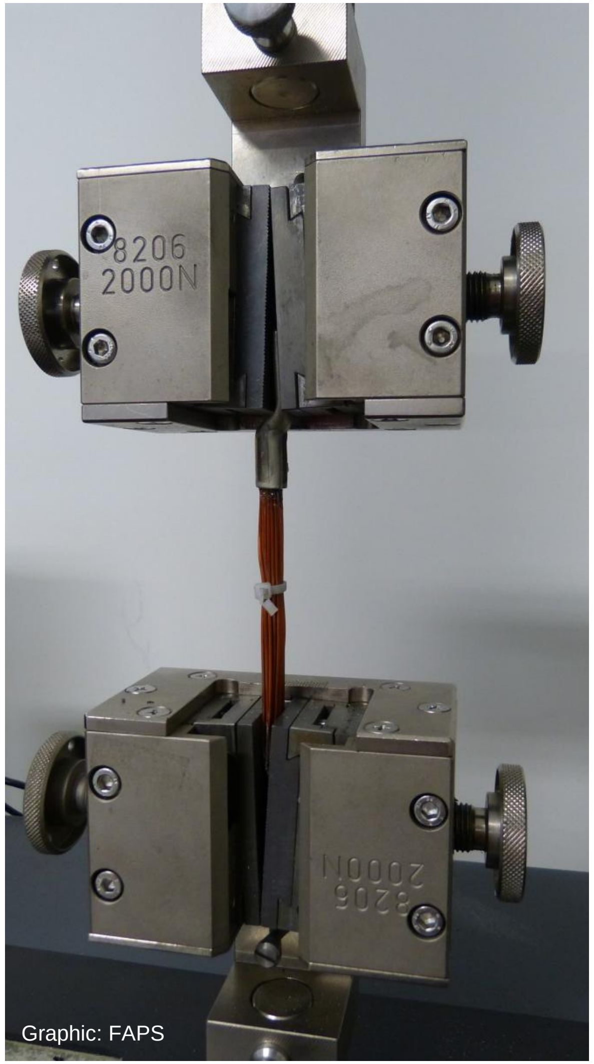
The qualification of the mechanical joint quality can be carried out by means of pull-out tests, shear tensile tests or peel tests.

DIN EN 60512-16-4 specifies a rough setting for the pull-out test

- Attachment the test specimen must not damage the cable lug or falsify the result
- Test specimen must be attached in such a way that the tensile force acts in the axial direction
- Rate of the moving head of the tensile testing machine between 25 and 50 mm/min
- End of test when conductor is pulled out of crimp sleeve or wire / test specimen breaks

DIN EN 60352-2 specifies minimum pull-out forces for cross-sections up to 10 mm²

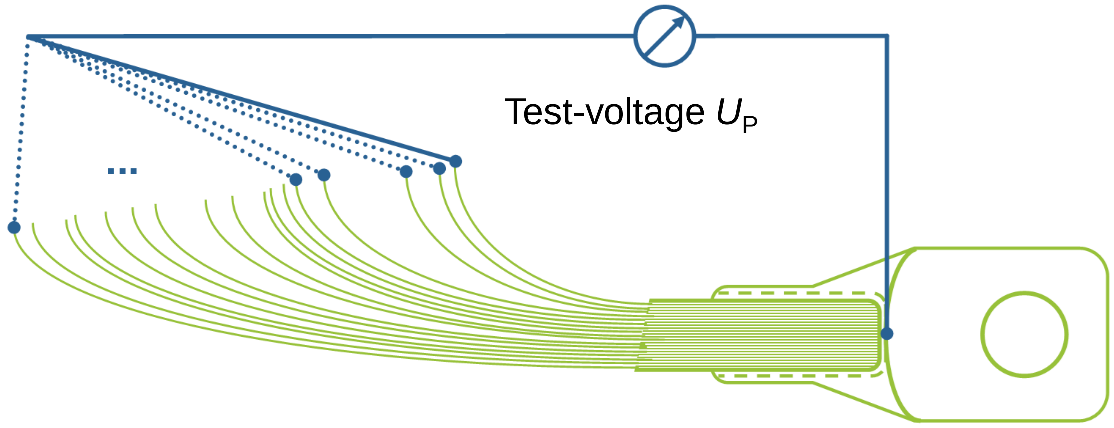
- Pull-out force: 0.05 mm² = 6 N, 10 mm² = 380 N
- Curve is not linear, therefore cannot simply be extrapolated linearly for larger cross-sections
- Additional measurement of individual pull-out forces as required, especially if the minimum pull-out forces are significantly exceeded



Graphic: FAPS

A quick and simple method for checking the electrical connection quality is to carry out a continuity test.

Continuity test
■ Application of a low test-voltage to check the electrical contact of the individual conductors to the e.g. crimped cable lug ■ Full contact as a "minimum requirement" for further measurement methods ■ Simple and quick to perform (depending on the number of individual conductors) ■ No quantitative statement possible

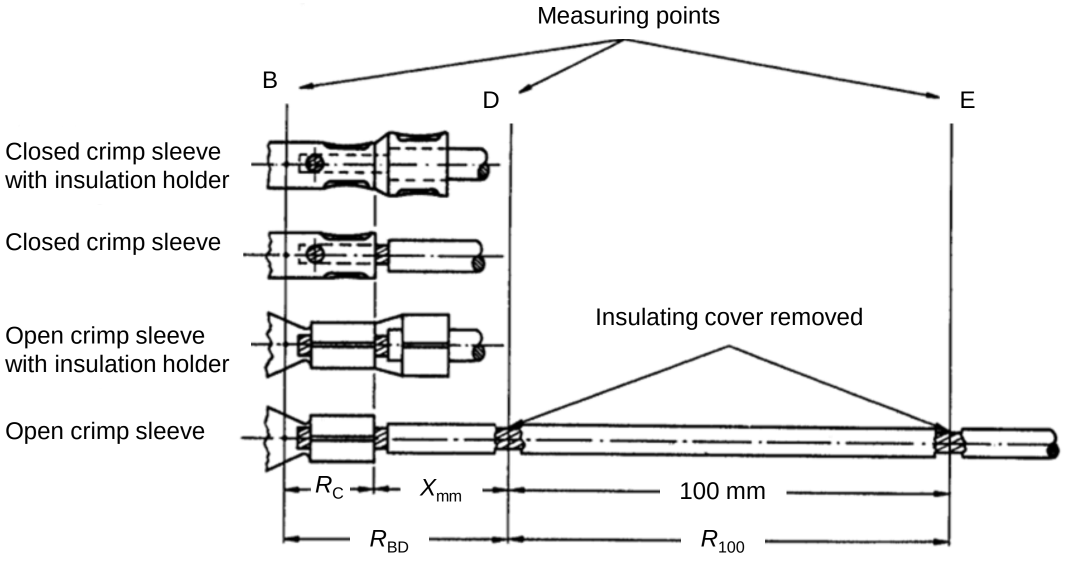


Graphic: FAPS

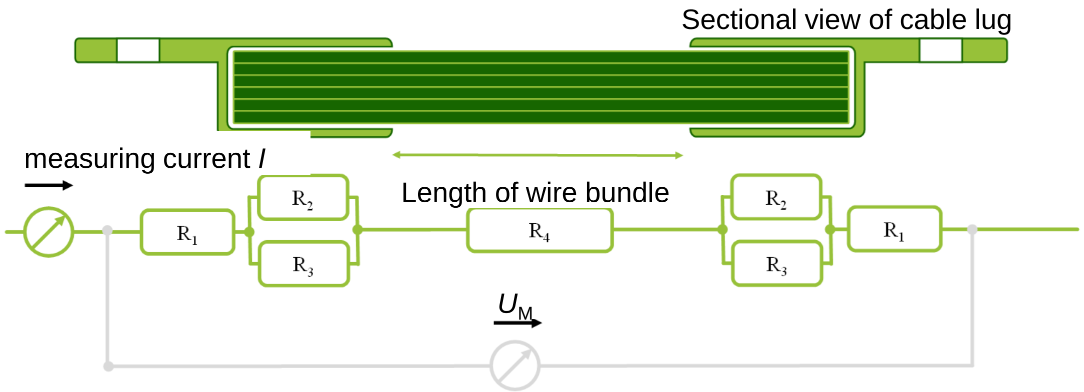
The measurement of contact resistance between two contact points enables a quantitative statement to be made about the electrical connection quality.

Determination of contact resistances

- Methods defined in standards only applicable for non-insulated strands
- Derivation of suitable measurement methods by setting up equivalent diagrams for the respective contacting case
- Measurement by energizing the samples and tapping the voltage
- Measured resistances usually in the range of a few µΩ



Graphic: DIN EN 60352-2



$$R_C = R_{BD} - \frac{X}{100} \cdot R_{100}$$

with:

- R_C – Contact resistance of the crimp connection in Ω
- R_{DB} – Measured resistance between B and D in Ω
- R_{100} – Measured resistance of 100 mm conductor length (D-E) in Ω
- X – Distance between crimp sleeve end and measuring points D in mm
- Note: 25 mm - 100 mm are recommended for the distance X

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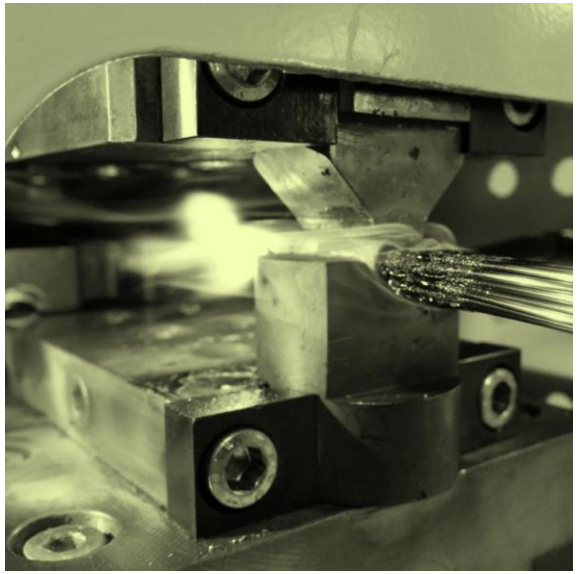
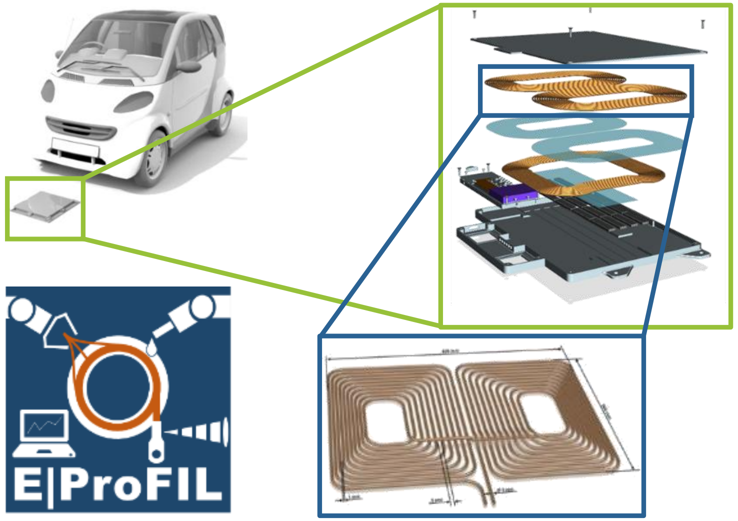


In the E|ProFIL project, the complete process chain of coil systems for inductive charging systems is being investigated and potentials of digitalization technologies are analysed.

E|ProFIL: Efficient processes for the production of inductive charging systems

Product design

- System design suitable for manufacturing
- Maximization of system efficiency
- Minimization of installation space requirements



Semi-finished product desing

- Processing-optimized strand design
- Processing-optimized fixation concepts

Process design

- Laying technologies
- Contacting technologies
- Insulation and fixing technologies
- Digitizing technologies

Gefördert durch:

Bundesministerium für Wirtschaft und Energie

DLR Projektträger

FAPS | zollner | LEONI | BOSCH Technik fürs Leben

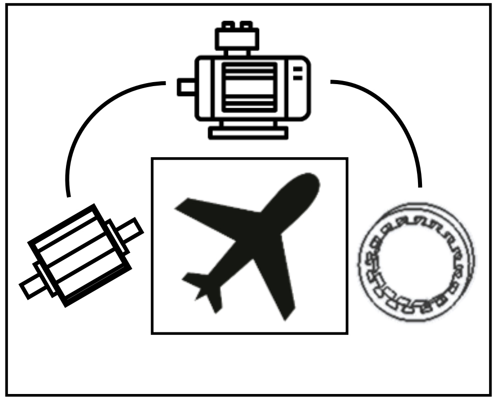
The research project TELEM addresses innovative processes for the production of efficient electric propulsion systems for aviation sector.

TELEM: Technological enabling of hybrid-electric propulsion systems for manned aircraft by researching aviation-grade electric machines and the integration into propulsion systems

Production and testing technologies for the electric propulsion system

Rotor production

- Lightweight materials and structures
- Assembly of high coercive segmented magnetic poles
- Innovative magnetic testing methods



Stator production

- Processing of soft magnetic materials
- Analysis of insulation and contacting processes
- Concept development for automated winding production

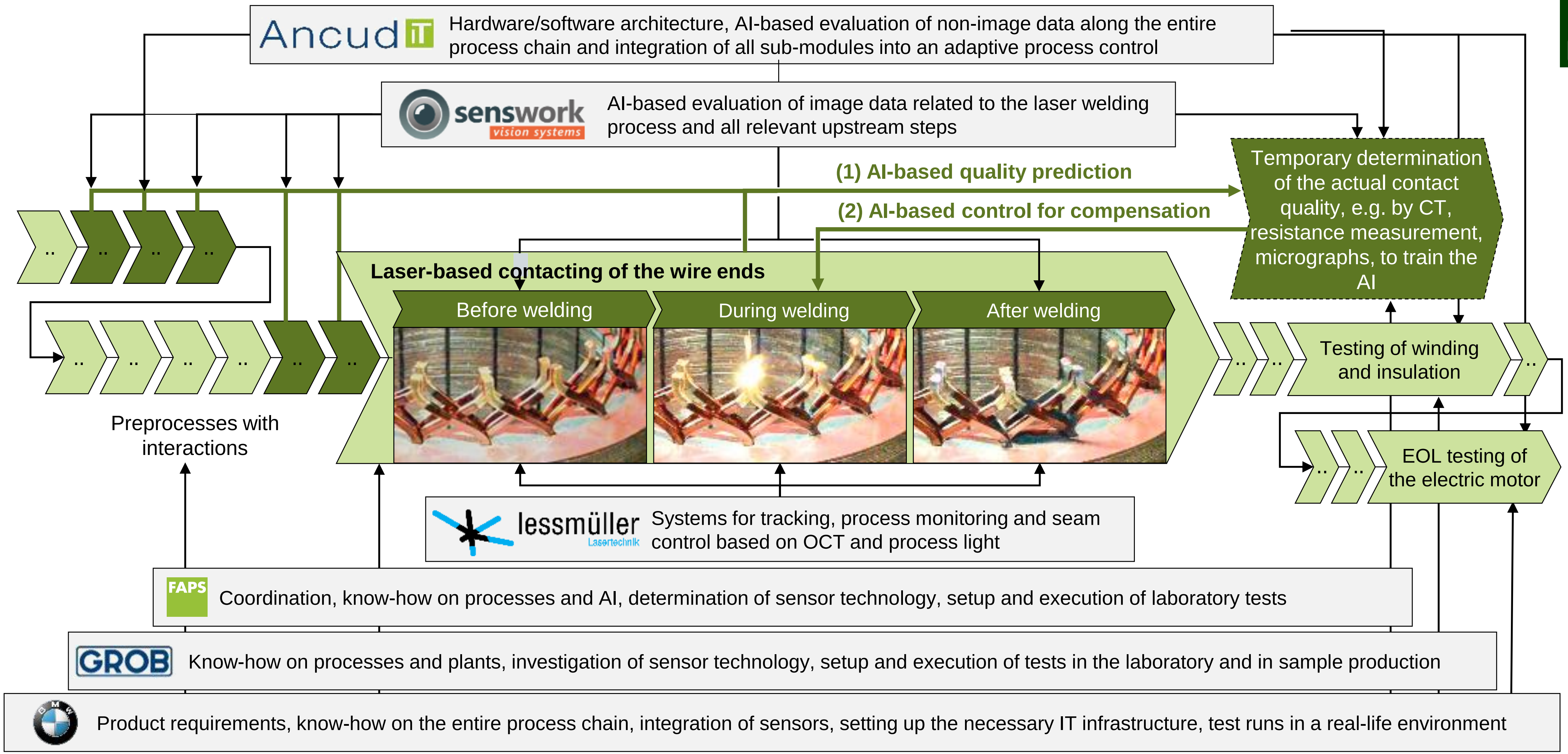
Full electric motor production

- Enhancement of major joining processes of full assembly
- End-of-line testing strategies

Gefördert durch:
Bundesministerium für Wirtschaft und Energie
DLR Projektträger

VAC VACUUMSCHMELZE
FAPS
ROLLS ROYCE
MTU Aero Engines
IABG
DLR Deutsches Zentrum für Luft- und Raumfahrt

The project KIKoSA investigates whether the quality of welded joints can be monitored by means of AI based on existing material and process data as well as easily integrable sensors.



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The contents of the lecture unit can be further deepened with the following specialized literature:

Books

- Katzier, H. (2015): „Elektrische Kabel und Leitungen: Technologien, Anwendungen und Anforderungen“, Leuze Verlag, Bad Saulgau.
- Katzier, H. (2012): „ Elektrische Steckverbinder : Technologien, Anwendungen und Anforderungen“, Leuze Verlag, Bad Saulgau.
- Mroczkowski, R. S. (2016): „Trilogie der Steckverbinder: Applikationshandbuch zur optimierten Steckverbinderauswahl“, Swiridoff Verlag, Künzelsau.
- Wodara, J. (2004): „Fachbuchreihe Schweißtechnik Band 151: Ultraschallfügen und –trennen“, DVS-Verlag, Düsseldorf.
- Holm, R. Electric Contacts. Theory and Application. Fourth completely rewritten edition. Berlin: Springer, 1967. ISBN 978-3-642-05708-3
- Slade, P.G., Hg. Electrical contacts. Principles and applications. 2. ed. Boca Raton, Fla.: CRC Press, 2014. ISBN 978-1-4398-8131-6
- Feldmann, K., V. Schöppner und G. Spur, Hg. Handbuch Fügen, Handhaben, Montieren. 2., vollständig neu bearbeitete Auflage. München: Hanser, 2014. Edition Handbuch der Fertigungstechnik. / hrsg. von Günter Spur ; 5. ISBN 978-3-446-42827-0
- Tzscheutschler, R., H. Olbrich und W. Jordan. Technologie des Elektromaschinenbaus. Berlin: Verl. Technik, 1990. ISBN 3-341-00851-9

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The contents of the lecture unit can be further deepened with the following specialized literature:

Papers

- Gläßel, T., J. Seefried, A. Kühl und J. Franke. Skinning of Insulated Copper Wires within the Production Chain of Hairpin Windings for Electric Traction Drives. In: International Journal of Mechanical Engineering and Robotics Research, S. 163-169
- Gläßel, T. und J. Franke. Kontaktierung von Antrieben für die Elektromobilität [online]. ZWF Zeitschrift für wirtschaftlichen Fabrikbetrieb, 2017, 112(5), S. 322-326. ISSN 0947-0085. Verfügbar unter: doi:10.3139/104.111721
- Gläßel, T., J. Seefried und J. Franke. Challenges in the manufacturing of hairpin windings and application opportunities of infrared lasers for the contacting process. In: *2017 7th International Electric Drives Production Conference (EDPC):* IEEE, 5. Dezember 2017 - 6. Dezember 2017, S. 1-7. ISBN 978-1-5386-1069-5
- Seefried, J., T. Gläßel, A. Kühl, A. Mayr und J. Franke. Experimental Evaluation of Tool Geometries for the Ultrasonic Crimping Process for Tubular Cable Lugs. In: *Proceedings of the Sixty-Fifth IEEE Holm Conference on Electrical Contacts.* Piscataway, NJ: IEEE, 2019, S. 178-183. ISBN 978-1-7281-3684-4

Dissertations

- Spreng, S.: *Numerische, analytische und empirische Modellierung des Heißcrimpprozesses.* Dissertation, Erlangen, 2020.
- *Gläßel, T.: Prozessketten zum Laserstrahlschweißen von flachleiterbasierten Formspulenwicklungen für automobile Traktionsantriebe. Dissertation, Erlangen, 2021.*

The contents of the lecture unit can be further deepened with the following specialized literature:

Graphics

- ATOP S.p.A.. Automotive [online]
- Aumann AG. Leistungen. Branchenlösungen. E-Mobility & Automotive. Elektrischer Antrieb. Hairpin Stator [online]
- bdtronic GmbH. Imprägniertechnologie. Anwendungsbereiche der Imprägniertechnologie [online]
- CTS GmbH. Produkte. Schock tss [online]
- ERICH GRAU GMBH. Produkte. Elektromotorenbleche/-pakete [online]
- Gustav Klauke GmbH. Produkte. Elektro. Pressen. Elektromechanische Crimpzange [online]
- Klaus Faber AG. Kabelschuhverpressung [online]
- risomat GmbH & Co.KG. Produkte. Prozesse [online]
- Rudolf Pack GmbH & Co. KG. Produkte. Hochfrequenzlitze. Rupalit-Profil [online]
- WITTENSTEIN cyber motor GmbH. Produkte. Servomotoren. Rotative-synchronmotoren. Cyber-kit-motor-gehaeuselose-servomotoren. Kabelabgang
- ZwickRoell GmbH & Co. KG. Branchen. Metall. Flachprodukte. Bleche und Bänder [online]

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

Institute for Factory Automation and Production Systems

Friedrich-Alexander-Universität
Technische Fakultät

THANK YOU